

DEEP REVIEW



Damla Başak Uçar 22301664

Zeynep Özdemir 22303314

Esra Nur Tat 22301880

Selin Demirci 22303722

Semih Çağlar 22303186

Zeynep Sude Kaplan 22401640

1. Introduction / Project Overview	2
2. High-Level System Structure / Architecture Overview	3
3. Detailed Build Instructions	4
4. Design Patterns Used	9
5. Core Features	12
6. Additional Features	13
7. User Manual	14
8. Work Allocation.....	43

1. Introduction / Project Overview

The *Deep Review* is a full-stack academic paper review management application developed to support the organization of review workflows in a lab-based environment. Based on the repository structure and implementation artifacts, the system brings together several major functions, including user signup and login, account handling, lab-oriented organization, paper registration, review round management, reviewer assignment, reviewer responses, coordinator approval and oversight processes, administrative management, and AI-related backend support components.

The main problem addressed by the system is the coordination of academic paper review activities within a centralized software environment. In practice, academic review workflows require multiple connected operations, such as user management, paper submission, reviewer assignment, review progress tracking, and coordinator supervision. The *Deep Review* provides an integrated application structure for managing these related activities through a shared backend and frontend implementation, thereby improving consistency and reducing fragmentation across the review process.

The primary user roles clearly visible in the current codebase are **LabMember**, **Coordinator**, and **Admin**, as defined in `backend/src/entities/User.ts`. These roles indicate that the system distinguishes between users who participate in lab and review workflows, users who coordinate approval and review management processes, and users who perform system-level administrative tasks. Although the repository also contains a `LocalAdmin.ts` entity file, the stable role enum visible in `User.ts` defines **LabMember**, **Coordinator**, and **Admin**; therefore, these three roles are the ones described in this report section.

2. High-Level System Structure / Architecture Overview

The repository is organized as a monorepo with a clear separation between the frontend and backend applications. The backend implementation is located in `backend/`, while the frontend implementation is located in `frontend/`. In addition, the repository contains supporting directories such as `test/`, `doc/`, `assumptions/`, `docs/`, and `Deliverables/`, which store tests, documentation, assumptions, design-related artifacts, and project deliverables.

The backend is implemented as a TypeScript-based Express application. The main application setup is defined in `backend/src/app.ts` and `backend/src/index.ts`, while central route registration is handled in `backend/src/routes/index.ts`. The backend follows a layered organization in which routes define the API surface, controllers handle request processing, services encapsulate reusable backend logic, entities define persistence models, and middleware provides authentication and access-control functionality.

The controller layer is organized according to major system responsibilities. The repository includes controllers such as `AccountController.ts`, `AdminController.ts`, `PaperController.ts`, `RoundController.ts`, `AssignmentController.ts`, `ReviewerResponseController.ts`, `RatingAnalyticsController.ts`, `TopicController.ts`, `SearchController.ts`, and `AIController.ts`.

These controllers correspond to functional areas including account operations, administration, paper handling, review round management, reviewer assignment, reviewer responses, analytics, topic management, search, and AI-related backend operations.

Reusable backend logic is placed in the service layer under `backend/src/services/`. Representative service files include `PaperService.ts`, `RoundService.ts`, `CoordinatorService.ts`, `ReviewerResponseService.ts`, `emailService.ts`, `tokenService.ts`, and `accountSecurity.ts`. The persistence layer is represented by TypeORM entities under `backend/src/entities/`. Core entities include `User.ts`, `Lab.ts`, `Paper.ts`, `Round.ts`, `Assignment.ts`, `ReviewerResponse.ts`, and `Topic.ts`, together with additional entities for notifications, ratings, analytics, templates, policies, audit logs, and related backend artifacts. Database setup is handled in `backend/src/data-source.ts`, and authentication middleware is defined in `backend/src/middleware/auth.ts`.

The frontend is implemented as a Next.js application using the App Router structure. Route pages are located under `frontend/src/app/`, including pages for login, signup, dashboard, papers, paper detail, registration, assigned reviews, lab members, pending approvals, and administration. This structure places the main user-facing workflows into route-specific page components.

Shared frontend layout and navigation are managed through files under `frontend/src/components/layout/`. The global layout is defined in `frontend/src/app/layout.tsx`, while `AppLayout.tsx` and `Sidebar.tsx` provide the main application shell and navigation structure. Profile-related user interface functionality is represented by `frontend/src/components/layout/ProfileModal.tsx`. User state is managed through React context in `frontend/src/components/context/UserContext.tsx`.

Communication with the backend is centralized through frontend API utilities in `frontend/src/lib/api.ts`. Authentication-related browser storage and user-mapping utilities are located in `frontend/src/lib/auth.ts`, while configuration support appears in `frontend/src/lib/config.ts`. The file `frontend/src/lib/mockData.ts` is present as legacy or mock support. The frontend and backend are connected through REST-style API routes, with backend route definitions in `backend/src/routes/index.ts`, frontend API helper functions in `frontend/src/lib/api.ts`, and proxy/configuration support visible in `frontend/next.config.mjs`.

Overall, the current repository reflects a role-aware web application architecture composed of a TypeScript Express backend with TypeORM-based persistence and middleware-based access control, paired with a Next.js frontend that organizes user workflows through route pages, shared layout components, React context, and centralized API/authentication utilities.

3. Detailed Build Instructions

Before you begin, ensure you have the following installed on your system:

Node.js (v16 or higher) - [Download](#)

npm (comes with Node.js)

System Requirements

RAM: Minimum 4GB (8GB recommended)

Disk Space: Minimum 1GB

OS: Windows, macOS, or Linux

Project Structure

Team8/

```
├── backend/ # Node.js/Express server
│   ├── src/
│   │   ├── controllers/ # API endpoint handlers
│   │   ├── services/ # Business logic
│   │   ├── entities/ # Database models (TypeORM)
│   │   ├── routes/ # API routes
│   │   ├── ai_content/ # AI features
│   │   │   ├── services/ # AI service implementations
│   │   │   ├── prompts/ # AI prompt templates
│   │   │   └── secrets.yaml # API keys and credentials
│   │   ├── types/ # TypeScript type definitions
│   │   ├── utils/ # Helper utilities
│   │   └── index.ts # Server entry point
│   ├── .env # Environment variables
│   ├── .env.example # Environment template
│   ├── package.json # Dependencies
│   └── tsconfig.json # TypeScript configuration
├── frontend/ # Next.js React application
│   ├── src/
│   │   ├── app/ # Page routes
│   │   ├── components/ # React components
│   │   ├── lib/ # Utilities and API client
│   │   └── styles/ # CSS/styling
│   └── package.json # Dependencies
```

```
| └─ next.config.js # Next.js configuration
|
└─ README.md # Project overview
```

Environment Setup

Backend Environment Variables (.env)

Create a .env file in the backend/ directory. Use .env.example as a template.

Required Variables

JWT_SECRET: Generate a secure random string (minimum 32 characters) for authentication

FRONTEND_BASE_URL: Where your React app is running (default: <http://localhost:3000>)

SMTP_HOST: smtp.gmail.com

SMTP_PORT: 587

SMTP_USER: Your Gmail address

SMTP_PASS: Gmail App Password (not your main password)

SMTP_FROM: Your Gmail address

OPENALEX_MAILTO (optional): Your email for OpenAlex “polite pool” access

AI Content Secrets Configuration

The AI features require API credentials stored in backend/src/ai_content/secrets.yaml.

Important: This file contains sensitive credentials. Never commit it to public repositories.

secrets.yaml Structure

Azure OpenAI Configuration

AZURE_OPENAI_ENDPOINT: "<https://your-resource.cognitiveservices.azure.com/>"

AZURE_OPENAI_KEY: "your_api_key_here"

AZURE_OPENAI_DEPLOYMENT: "gpt-5.4-mini"

AZURE_OPENAI_MODEL: "gpt-5.4-mini"

AZURE_OPENAI_API_VERSION: "2024-12-01-preview"

How to Obtain These Credentials

Azure Tenant ID & Client ID:

Log in to Azure Portal

Navigate to Azure Active Directory → App registrations

Create or select your application

Copy Directory (tenant) ID and Application (client) ID

Azure Client Secret:

In the same app registration, go to Certificates & secrets

Create a new client secret

Copy the value immediately (it won't be shown again)

Azure OpenAI Credentials:

Navigate to Azure OpenAI resource in Azure Portal

Go to Keys and Endpoint

Copy the API key and endpoint URL

AI Agent IDs:

Go to Azure AI Foundry (<https://ai.azure.com>)

Create or locate your agents

Each agent has a unique ID starting with asst_

Installation Steps

1. Clone the Repository

```
git clone https://github.com/CS319-26-SP/Team8.git  
cd Team8
```

2. Install Backend Dependencies

```
cd backend  
npm install
```

3. Install Frontend Dependencies

```
cd ../frontend
npm install
cd ..
```

4. Database

The project uses SQLite with the database file already included in the repository. When you clone the project, the database is automatically available at backend/database.sqlite.

No additional setup is required - TypeORM automatically manages the schema.

5. Configure Environment Files

Backend:

```
cd backend
cp .env.example .env
# Edit .env with your actual values
nano .env # or use your preferred editor
```

AI Content Secrets:

```
# Place your secrets.yaml in backend/src/ai_content/
# The file should contain all Azure AI credentials
```

Database Configuration

The application uses SQLite with TypeORM as the ORM layer.

Database File: backend/database.sqlite (automatically created on first run)

Auto-sync: TypeORM automatically creates/updates tables based on entity definitions in development mode.

No manual setup required - the database is included in the repository and will be ready when you clone the project.

Running the Application

Terminal 1 - Start Backend Server

```
cd backend
npm run dev
```

Server will start on <http://localhost:3001>

Logs will show: 🚀 Server started on <http://localhost:3001>

Terminal 2 - Start Frontend Development Server

```
cd frontend  
npm run dev
```

Frontend will start on <http://localhost:3000>

Open your browser and navigate to <http://localhost:3000>

4. Design Patterns Used

The implementation uses several architectural and implementation patterns that are directly visible in the current codebase. These patterns contribute to the separation of responsibilities between request handling, business logic, persistence, shared frontend state, and reusable interface structure.

4.1 MVC / Layered Architecture Pattern

The system follows a web-oriented MVC-style layered structure. The backend separates route definitions, controllers, services, and entities, while the frontend separates route pages, layout components, context, and API utilities.

Implemented in:

backend/src/routes/index.ts
backend/src/controllers/AccountController.ts
backend/src/controllers/PaperController.ts
backend/src/controllers/RoundController.ts
backend/src/services/PaperService.ts
backend/src/services/RoundService.ts
backend/src/entities/User.ts
backend/src/entities/Paper.ts
frontend/src/app/*
frontend/src/components/layout/AppLayout.tsx
frontend/src/lib/api.ts

Reason for use:

This structure separates responsibilities across the system. Routes define API endpoints, controllers handle HTTP request and response logic, services contain reusable business logic, and entities represent database models. On the frontend, pages represent user-facing screens, while shared layout components and API utilities reduce duplication and improve consistency.

4.2 Repository Pattern

The backend uses TypeORM repositories to access and modify database entities. Instead of writing raw SQL throughout the application, controllers and services access persistent data through repository objects.

Implemented in:

backend/src/data-source.ts
backend/src/controllers/AccountController.ts
backend/src/controllers/PaperController.ts
backend/src/services/PaperService.ts
backend/src/services/RoundService.ts

Reason for use:

The Repository Pattern abstracts database operations and keeps persistence access consistent. It makes entity operations such as finding, creating, updating, and saving records easier to manage through TypeORM and reduces the need for raw query logic throughout the codebase.

4.3 Singleton Pattern

The backend uses a single shared TypeORM data source instance for database access.

Implemented in:

backend/src/data-source.ts

Reason for use:

A single shared database connection/configuration object avoids repeatedly creating new data source instances. This centralizes database configuration and ensures that the backend uses one consistent connection setup.

4.4 Chain of Responsibility Pattern

The backend uses Express middleware chains for authentication and authorization. Requests pass through middleware functions before reaching the final controller method.

Implemented in:

backend/src/middleware/auth.ts

backend/src/routes/index.ts

Reason for use:

This pattern allows access-control responsibilities to be handled step by step. A request is first authenticated, then role-checked, and only then passed to the controller. This avoids duplicating authorization logic inside every controller method and keeps route protection consistent.

4.5 Facade Pattern

The frontend uses a centralized API utility file that hides the details of raw backend requests from page components.

Implemented in:

frontend/src/lib/api.ts

Reason for use:

Frontend pages do not need to manually construct every request, attach authentication tokens, parse responses, or repeat common error-handling logic. The API utility layer provides a simpler and more consistent interface for backend communication.

4.6 DRY Principle

The codebase applies the DRY (Don't Repeat Yourself) principle by centralizing repeated logic in shared modules.

Implemented in:

frontend/src/lib/api.ts

frontend/src/lib/auth.ts

backend/src/middleware/auth.ts

backend/src/services/accountSecurity.ts

backend/src/services/tokenService.ts

Reason for use:

Common logic such as API requests, token handling, authentication checks, password hashing, login lockout rules, and role checks is centralized instead of being reimplemented in multiple pages or controllers. This improves maintainability and reduces the likelihood of inconsistent behavior.

5. Core Features

The current implementation supports several core features that form the main functionality of the *Deep Review*. These features cover user management, paper handling, review coordination, reviewer workflows, administration, analytics, and AI-assisted review support.

5.1 User Account and Authentication Management

The system supports user signup, login, logout, authenticated profile access, profile updates, password changes, and password reset flows. It also includes account security mechanisms such as password hashing, reset tokens, and temporary login lockout after repeated failed attempts. These features provide the foundation for secure access to the platform.

5.2 Role-Based Access and Account Approval

The system distinguishes between **LabMember**, **Coordinator**, and **Admin** roles. Protected functionality is controlled through role-based access checks, and coordinator-facing approval workflows allow pending accounts to be approved or rejected. This ensures that sensitive operations are restricted to authorized users.

5.3 Lab and Member Management

The system includes a lab-based organizational structure through lab entities, member retrieval, topic-related management, and coordinator/admin management capabilities. Coordinators can view lab members and manage certain member states, while administrators have broader management authority. Since the system is designed to support multiple labs, this structure also helps separate users, papers, and review-related activities according to their

academic context. In this way, the platform provides clearer organizational boundaries and improves coordination across lab-specific workflows.

5.4 Paper Registration and Paper Management

The system supports academic paper registration, paper listing, paper detail viewing, paper history, and updates to paper metadata such as abstract, authors, topics, status, and Overleaf link. This allows papers to be created, updated, and tracked throughout the review lifecycle.

5.5 Review Round Management

The system supports review rounds for papers, including round creation, round approval, deadline editing, round detail updates, reviewer proposal, and round-status tracking. This functionality organizes the review process into structured and manageable phases.

5.6 Reviewer Assignment and Reviewer Response Workflow

The system supports assigning reviewers, sending invitations, accepting invitations, requesting decline or extension, completing reviews, cancelling assignments, reassigning reviewers, and updating assignment deadlines. This feature area covers both coordinator-side assignment management and reviewer-side participation in the review process.

5.7 Pending Approval and Coordinator Management Interfaces

The system includes coordinator-facing approval interfaces for reviewing pending signup requests and related management actions. These interfaces allow coordinators to process requests that require manual review before users become active in the system.

5.8 Administration

The system provides administrator-facing functionality for managing users, labs, policies, templates, and audit-related information. This supports system-level oversight and configuration tasks required for maintaining the platform.

5.9 Search, Topics, Ratings, and Analytics

The system includes support for search functionality, topic organization, reviewer rating, and analytics. These features help users find relevant information, manage topic-related data, and evaluate reviewer performance.

5.10 AI-Related Review Support

The codebase also includes AI-related support for AI review, compliance checking, checklist handling, empirical checklist answers, submission-rule extraction, and reference verification. These features extend the broader review workflow with AI-assisted operations and advanced review-support functionality.

6. Additional Features

6.1 Account Security and Password Recovery

The system includes account-security mechanisms beyond basic login. Passwords are hashed before storage, password reset tokens are supported, and repeated failed login attempts can temporarily lock an account. These features strengthen the security of user authentication and account recovery.

6.2 Lab Topics and User Interests

The codebase includes topic-related functionality for organizing labs, users, and papers. Users can manage interests, and lab topics can be retrieved or managed through backend routes. This provides additional structure for categorization and profile-based organization.

6.3 Freezing and Unfreezing Lab Members

Coordinator-facing account management includes the ability to freeze and unfreeze lab members. This provides a way to restrict certain actions for inactive or temporarily restricted users and gives coordinators additional control over lab participation.

6.4 Collaboration Invitations Between Labs

The system includes support for lab collaboration invitations around papers. Coordinators can send invitations, view invitations for a paper, and accept, reject, or cancel pending collaboration invitations. This extends the review environment by supporting coordination across different labs.

6.5 Paper and Review History

The system includes support for viewing paper history and review-related status information. This helps users and coordinators track changes and monitor review progress over time.

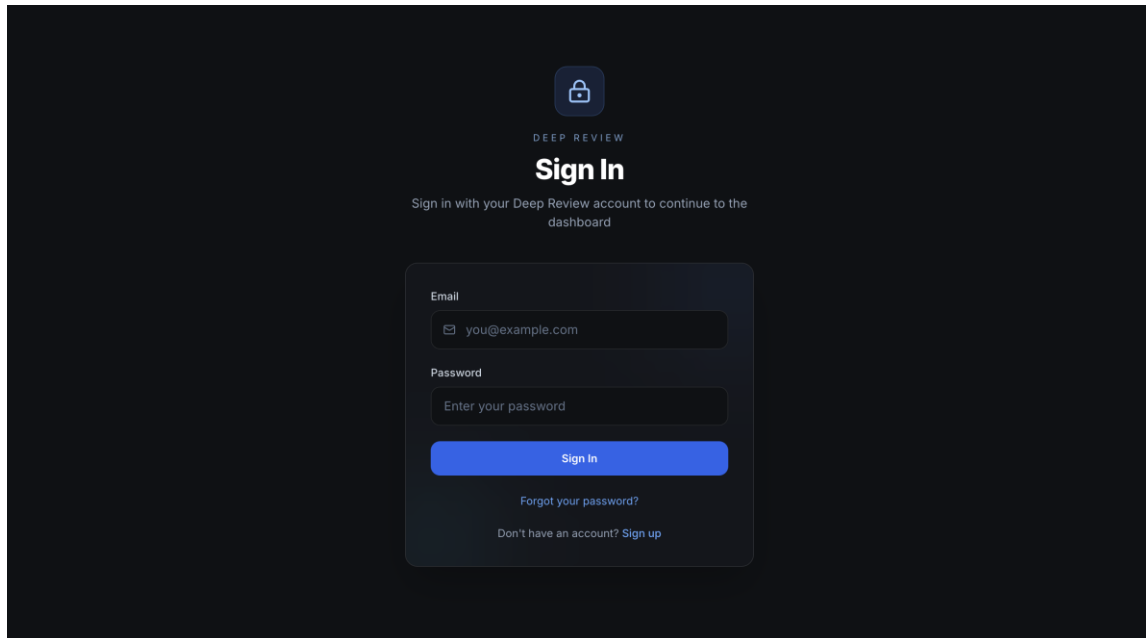
6.6 Citation Suggestion

The Citation Suggestion feature analyzes papers written by the target venue's committee members and suggests the ones most relevant to the submitted paper. It also shows relevance levels such as high, medium, or low, includes DOI links when available, verifies the results to reduce hallucinated references, and allows the suggestions to be downloaded as a single CSV file.

7. User Manual

Sign In & Sign Up

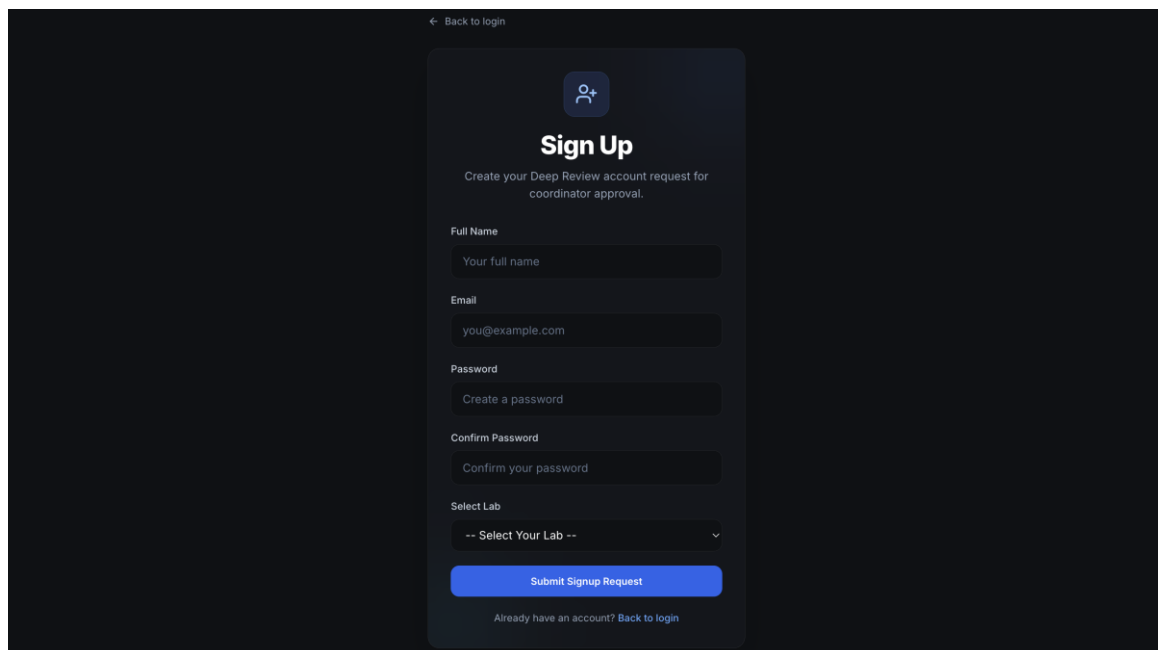
Sign In



The image shows a dark-themed 'Sign In' page for 'DEEP REVIEW'. At the top center is a lock icon. Below it, the text 'DEEP REVIEW' is in small caps, followed by 'Sign In' in a larger font. A subtitle reads: 'Sign in with your Deep Review account to continue to the dashboard'. The main form is a rounded rectangle containing two input fields: 'Email' with the placeholder 'you@example.com' and 'Password' with the placeholder 'Enter your password'. Below these is a blue 'Sign In' button. At the bottom of the form are two links: 'Forgot your password?' and 'Don't have an account? Sign up'.

Users sign in by entering their registered email address and password on the login page. After successful authentication, the system redirects them to the appropriate interface based on their role. If the credentials are incorrect, access is denied and an error message is shown.

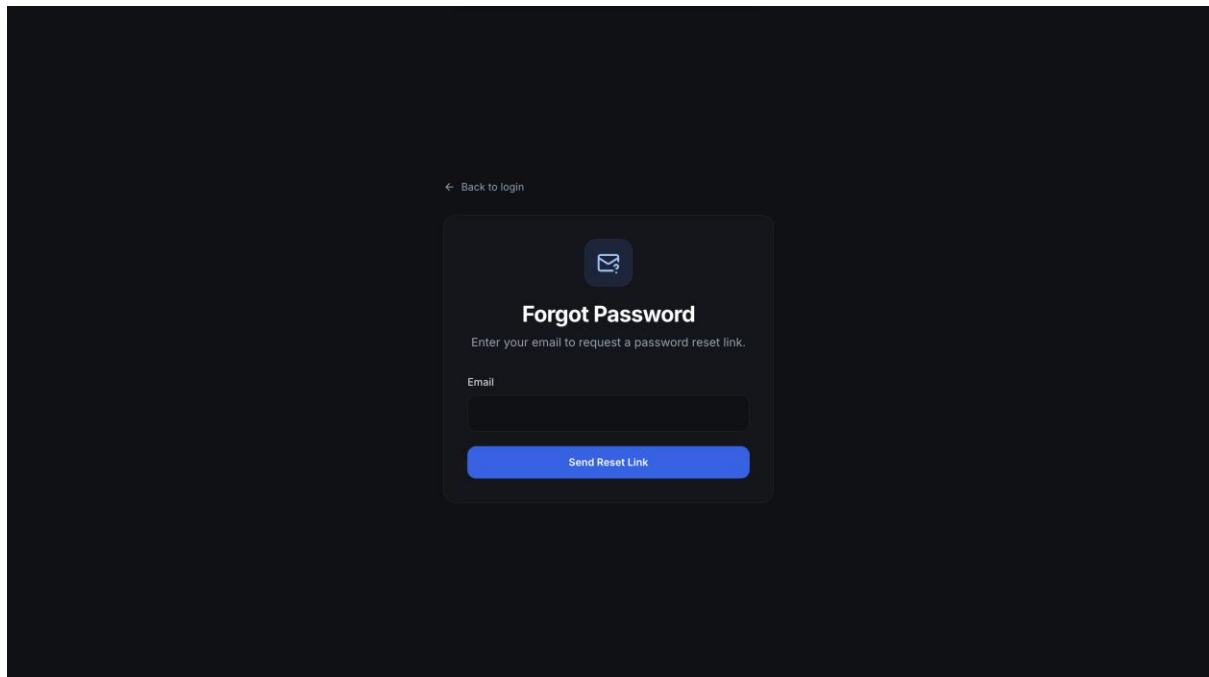
Sign Up



The image shows a dark-themed 'Sign Up' page for 'DEEP REVIEW'. At the top left is a '← Back to login' link. The main form is a rounded rectangle. At the top is a person icon with a plus sign, followed by 'Sign Up' in a large font. A subtitle reads: 'Create your Deep Review account request for coordinator approval.' The form contains five input fields: 'Full Name' (placeholder: 'Your full name'), 'Email' (placeholder: 'you@example.com'), 'Password' (placeholder: 'Create a password'), 'Confirm Password' (placeholder: 'Confirm your password'), and a 'Select Lab' dropdown menu (placeholder: '-- Select Your Lab --'). Below these is a blue 'Submit Signup Request' button. At the bottom of the form is a link: 'Already have an account? Back to login'.

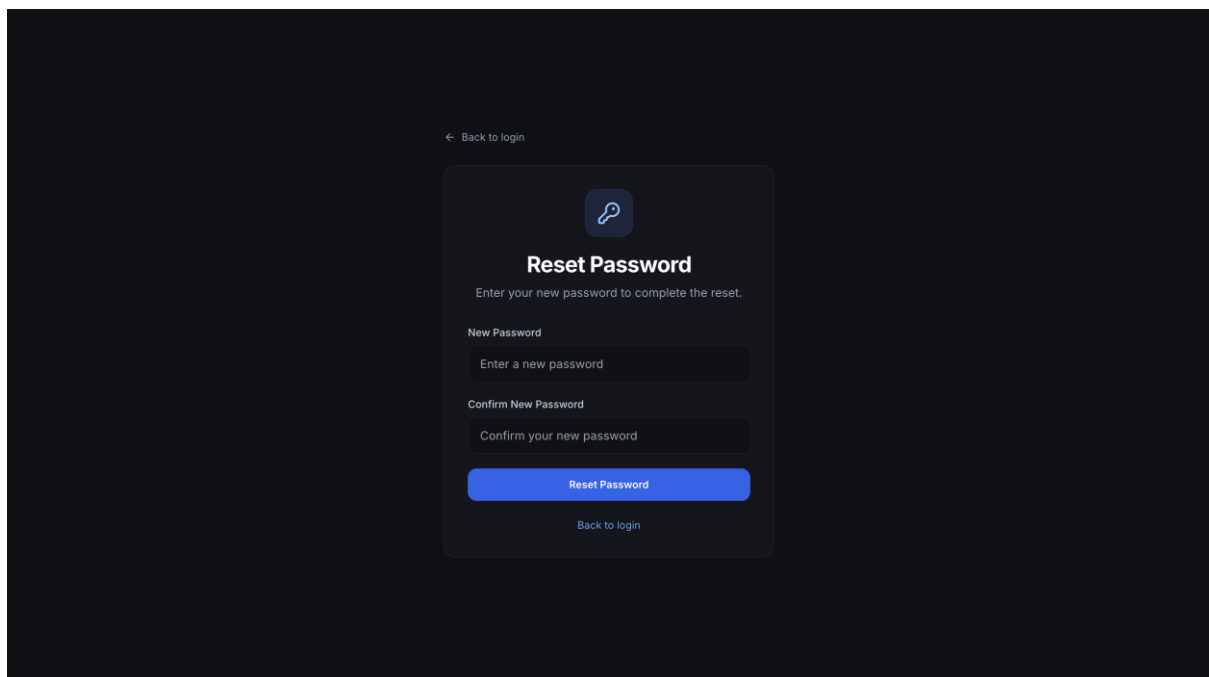
Users can create an account through the sign up page by providing the required personal and account information. After submission, the account remains pending until it is approved by a coordinator. Once the approval is completed, the user receives an approval email and can then sign in to the system with the assigned role.

Forgot Password



The screenshot shows a 'Forgot Password' form on a dark background. At the top left of the form is a link '← Back to login'. The form has a title 'Forgot Password' with an envelope icon above it. Below the title is the instruction 'Enter your email to request a password reset link.' followed by an 'Email' label and a text input field. At the bottom of the form is a blue button labeled 'Send Reset Link'.

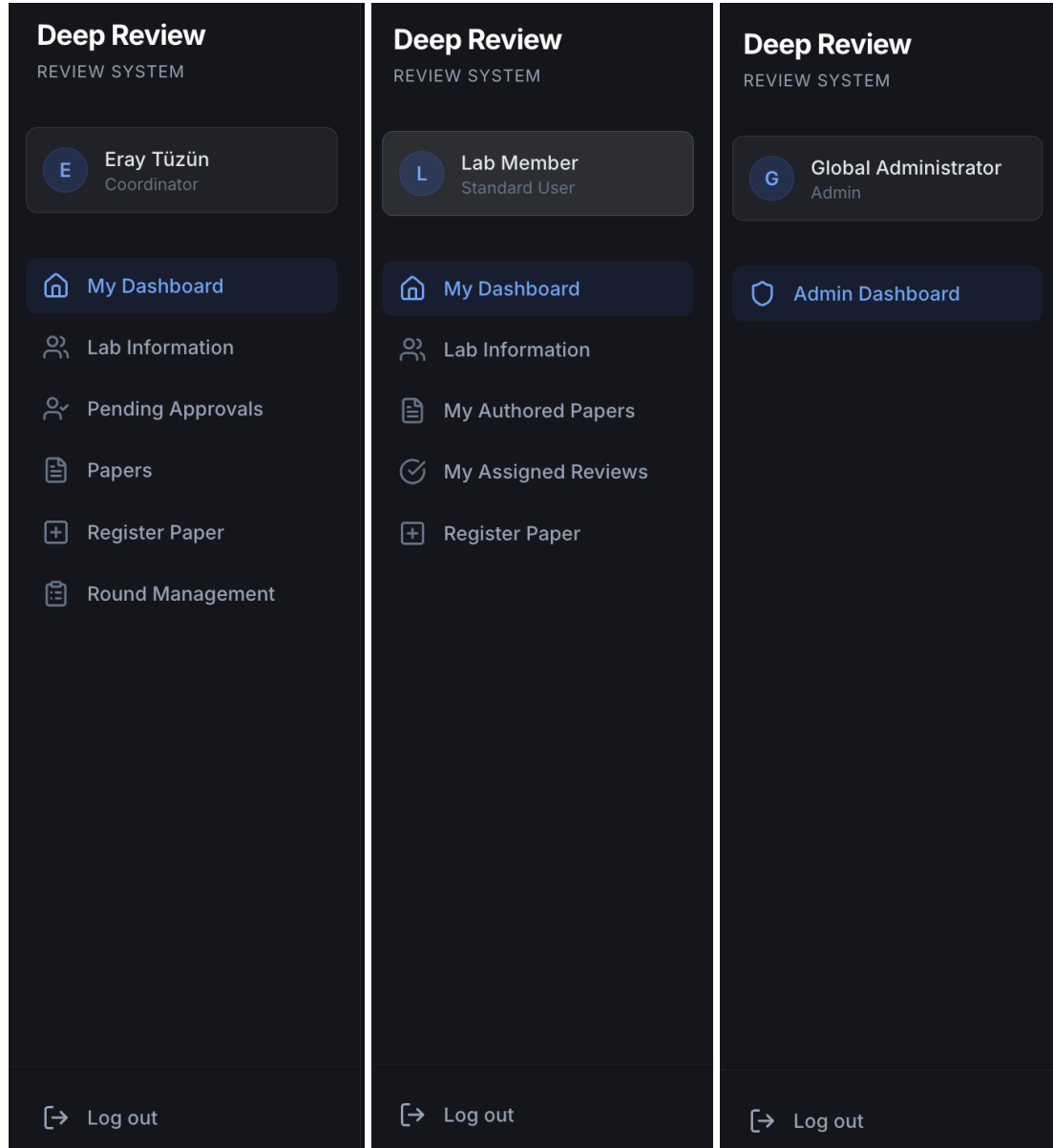
Users who forget their password can enter their registered email address on the Forgot Password page and request a reset link. The system then sends a password reset email to that address, allowing the user to create a new password and regain access to the account.



The screenshot shows a 'Reset Password' form on a dark background. At the top left of the form is a link '← Back to login'. The form has a title 'Reset Password' with a key icon above it. Below the title is the instruction 'Enter your new password to complete the reset.' followed by two sections: 'New Password' with an 'Enter a new password' label and text input field, and 'Confirm New Password' with a 'Confirm your new password' label and text input field. At the bottom of the form is a blue button labeled 'Reset Password' and a link 'Back to login' below it.

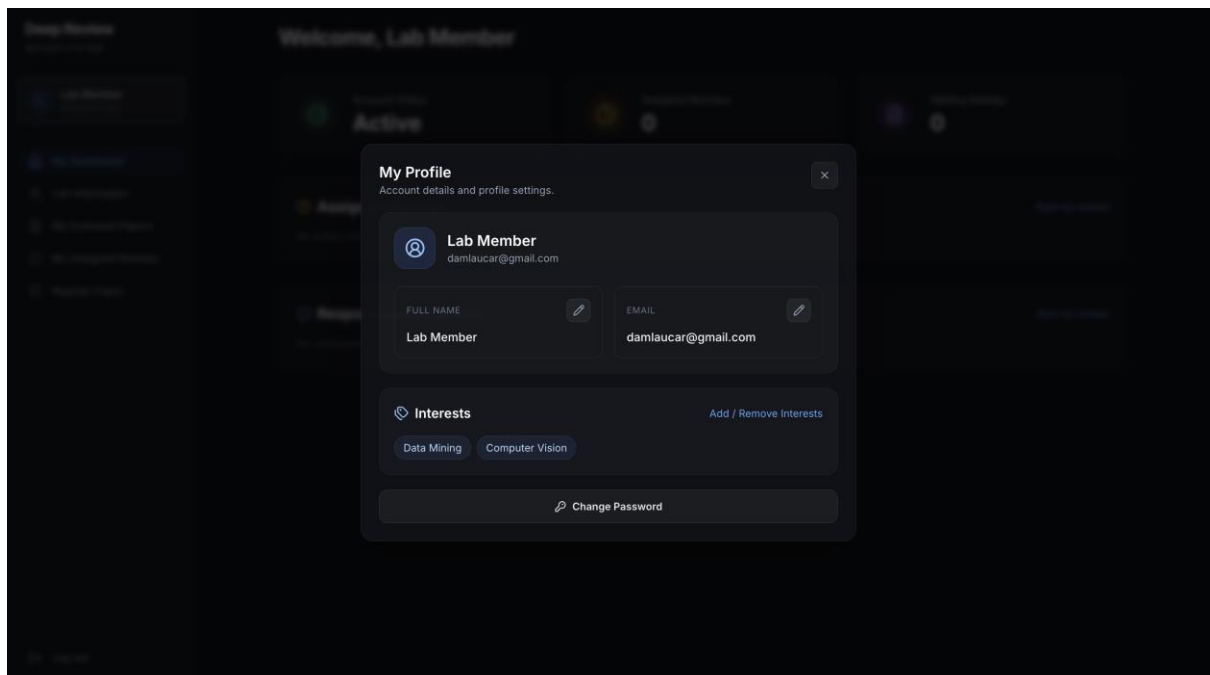
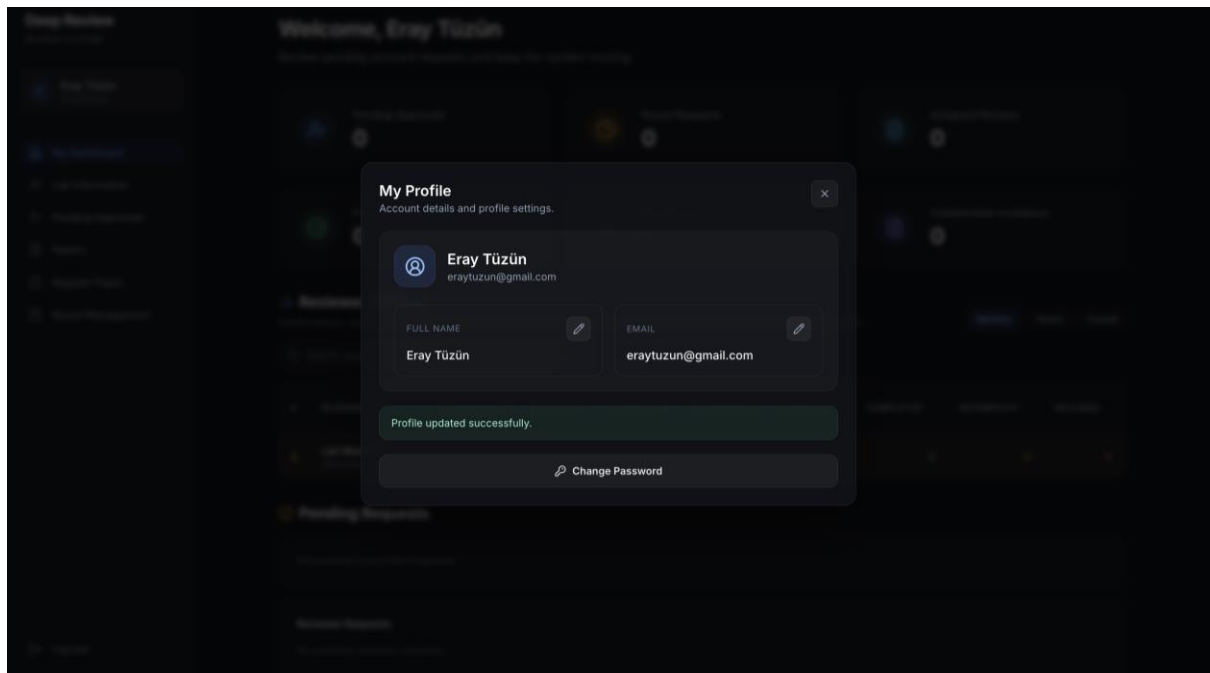
The Reset Password page allows users to create a new password after opening a valid reset link. The user must enter and confirm the new password, then click **Reset Password** to complete the process.

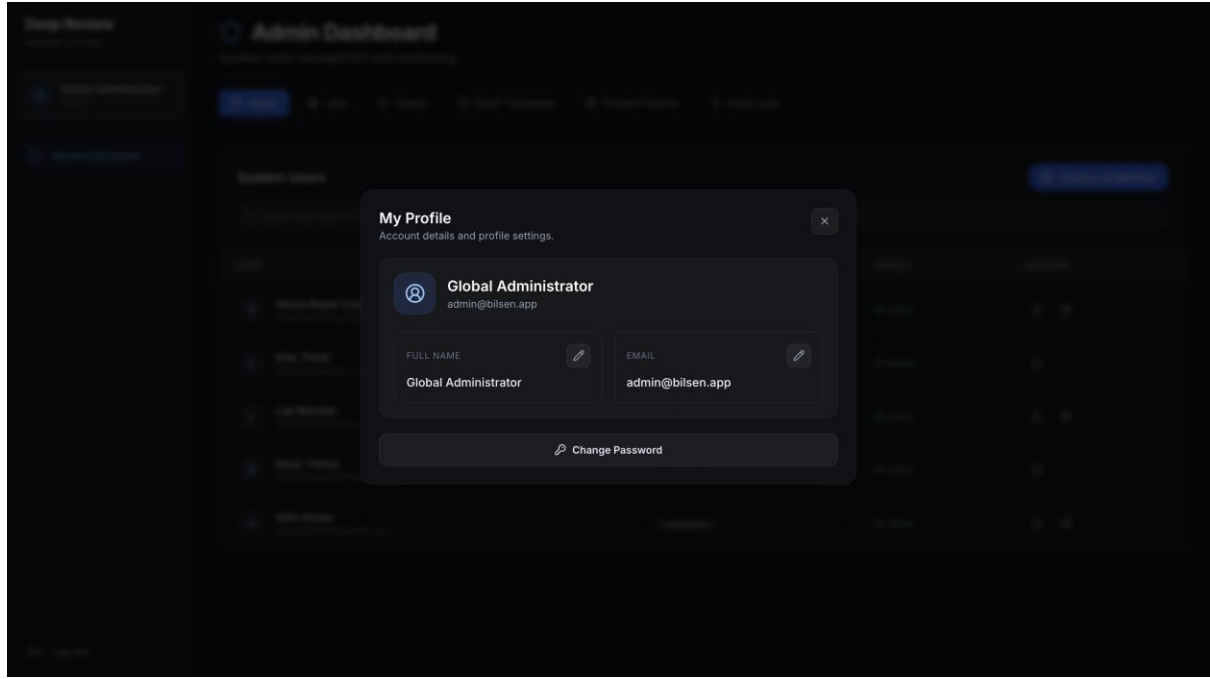
Sidebar



Sidebar provides role-based navigation for the Deep Review system. It displays the current user's name and role at the top, then shows only the menu options that the user is authorized to access, such as dashboards, paper registration, review assignments, approvals, and round management. This helps coordinators, lab members, and administrators quickly reach their relevant tasks without seeing unnecessary pages.

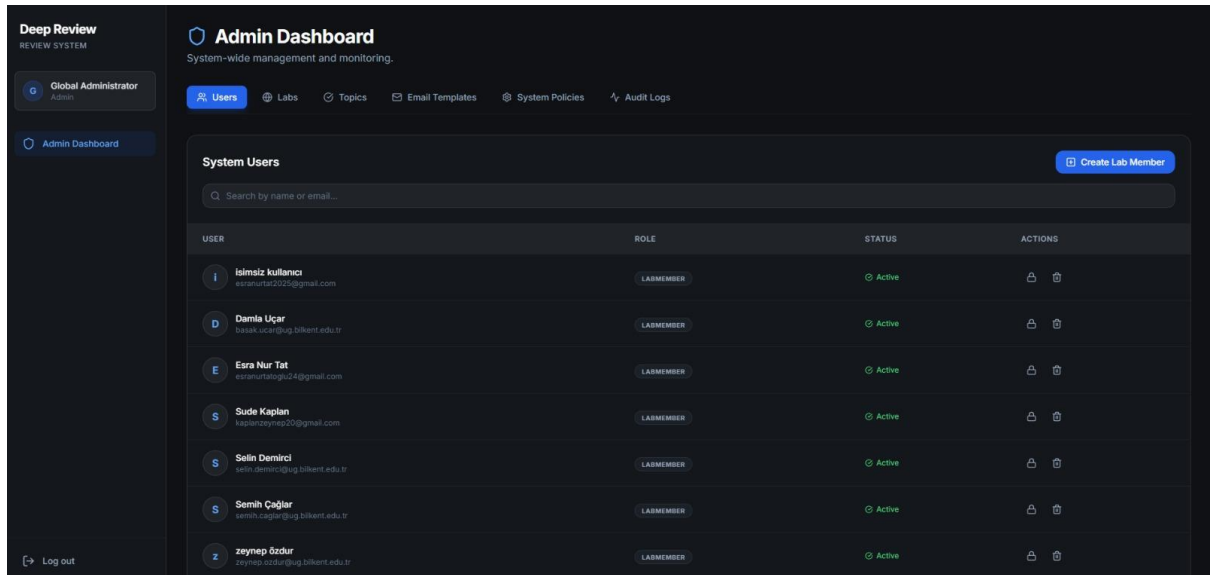
My Profile and Profile Update





My Profile and Profile Update section allows users to view and manage their account information. In the profile modal, users can see their name, email address, and role-based profile details. They can update editable personal information, change their password, and, for lab members, manage research interests used in the review assignment process.

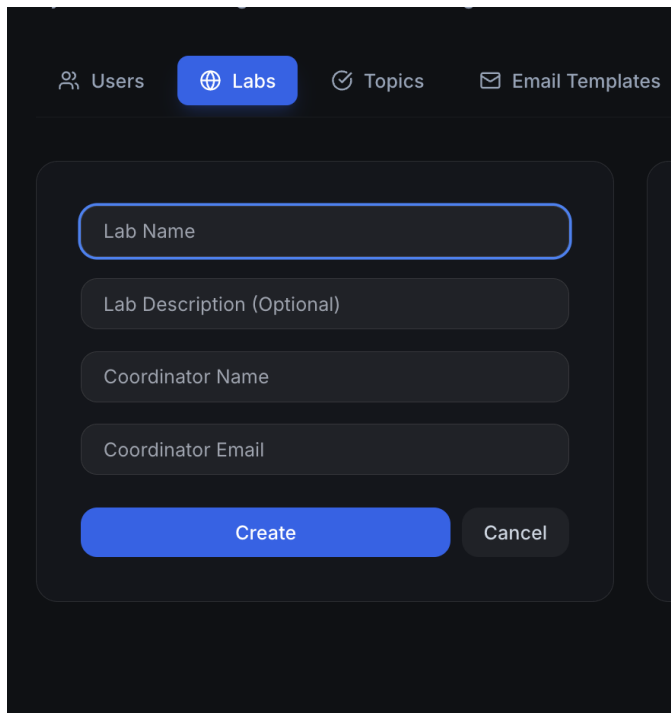
Admin Dashboard



Admin Dashboard allows the global administrator to monitor and manage system-level data. From this page, the admin can view registered users, check their roles and account statuses, and use separate tabs for labs, topics, email templates, system policies, and audit logs. The search function also helps the admin quickly find specific users by name or email.

In the Labs tab, the administrator can view existing labs and create a new lab. Each lab card shows the lab name, short description, and assigned coordinator.

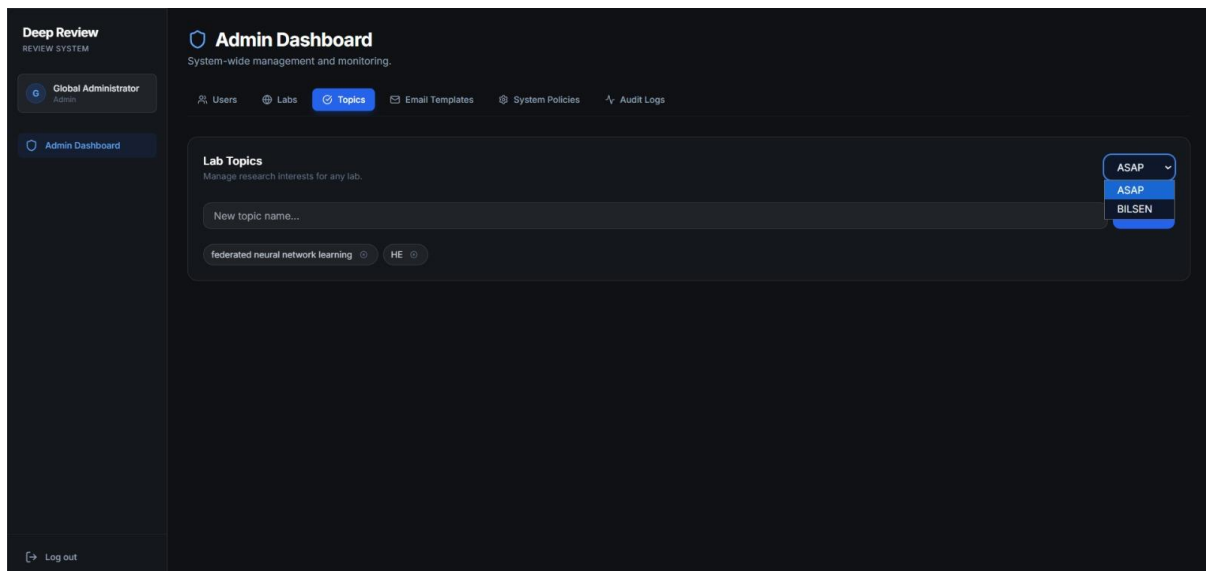
Admin Dashboard – Create Lab



The screenshot shows the 'Create Lab' form in the Admin Dashboard. The form is located under the 'Labs' tab, which is highlighted in blue. The form contains four input fields: 'Lab Name', 'Lab Description (Optional)', 'Coordinator Name', and 'Coordinator Email'. Below the input fields are two buttons: 'Create' (in blue) and 'Cancel' (in gray). The top navigation bar includes links for 'Users', 'Labs', 'Topics', and 'Email Templates'.

The administrator can create a new lab by entering the lab name, optional description, coordinator name, and coordinator email. After clicking **Create**, the new lab is added to the system with its assigned coordinator.

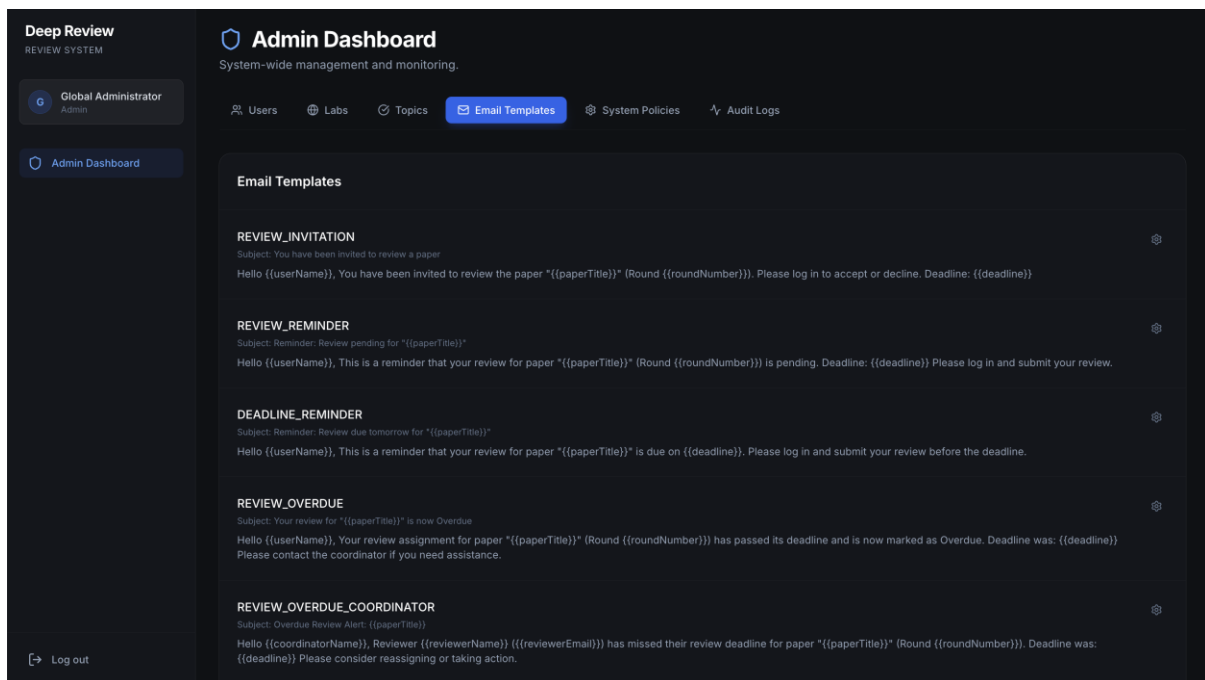
Admin Dashboard – Topics



The screenshot shows the 'Lab Topics' section in the Admin Dashboard. The 'Topics' tab is highlighted in blue. The section is titled 'Lab Topics' and includes the subtitle 'Manage research interests for any lab.' There is a dropdown menu on the right side with options 'ASAP', 'ASAP', and 'BILSEN'. Below the dropdown is a text input field labeled 'New topic name...'. At the bottom, there are two topic tags: 'federated neural network learning' and 'HE'. The left sidebar shows the 'Admin Dashboard' link. The top navigation bar includes links for 'Users', 'Labs', 'Topics', 'Email Templates', 'System Policies', and 'Audit Logs'.

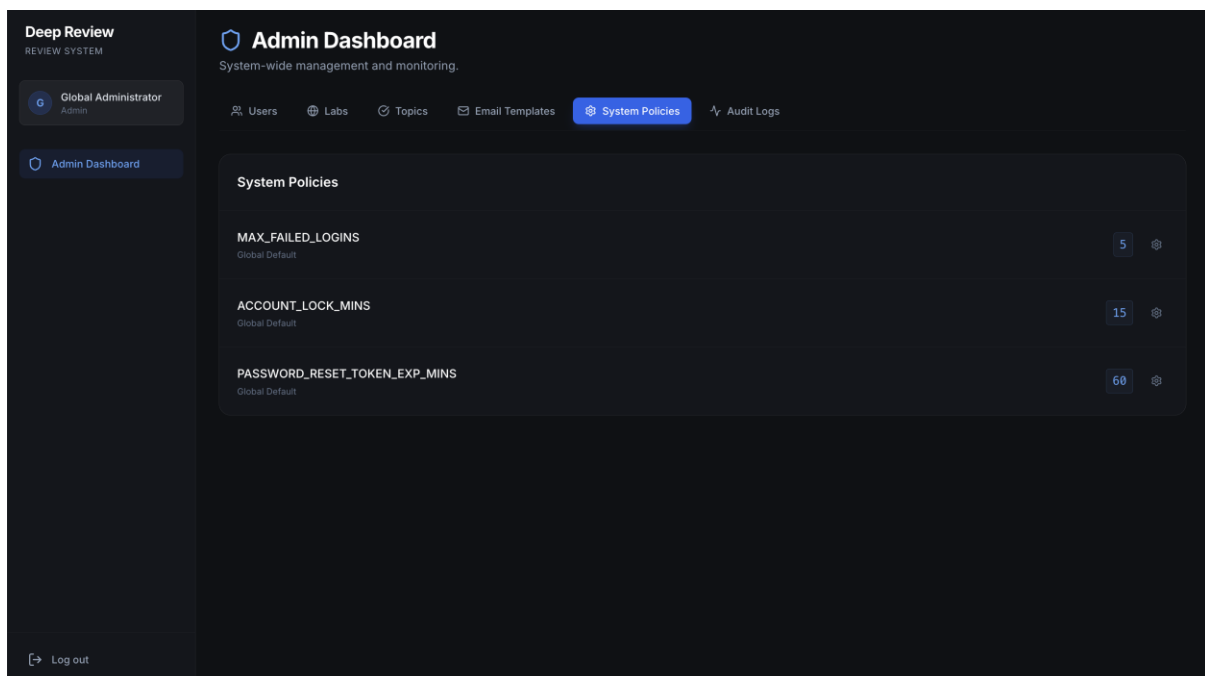
The Topics tab allows the administrator to manage research topic lists for each lab. The administrator can select a lab from the dropdown, add a new topic, and view or remove existing topics.

Admin Dashboard – Email Templates



The Email Templates tab shows the predefined email messages used by the system, such as review invitations, reminders, and overdue notifications. The administrator can review these templates and use the settings icon to manage template details.

Admin Dashboard – System Policies



The System Policies tab shows global security settings such as maximum failed login attempts, account lock duration, and password reset token expiration time. The administrator can review these policy values and use the settings icon to edit them when needed.

Admin Dashboard – Audit Logs

The screenshot shows the 'Admin Dashboard' with a sidebar on the left containing 'Global Administrator Admin' and 'Admin Dashboard'. The main content area is titled 'Audit Logs' and shows a table of system actions. The table has columns for 'TIME', 'ACTOR', 'ACTION', and 'DETAILS'. The actions listed are 'LOGIN SUCCESS' for various users, including 'Global Administrator', 'Semih Çağlar', 'zeynep özür', 'Eray Tüzün', 'İsimsiz kullanıcı', and 'Semih Çağlar'. The table is filtered by 'All actions' and sorted by 'Newest first'. The sidebar also includes a 'Log out' button.

TIME	ACTOR	ACTION	DETAILS
5/3/2026, 9:22:55 PM	Global Administrator ADMIN	LOGIN SUCCESS	User (43bf4bb4...)
5/3/2026, 8:02:23 PM	Semih Çağlar LABMEMBER	LOGIN SUCCESS	User (9993fea0...)
5/3/2026, 8:02:09 PM	zeynep özür LABMEMBER	LOGIN SUCCESS	User (5e55d16b...)
5/3/2026, 7:56:40 PM	Semih Çağlar LABMEMBER	LOGIN SUCCESS	User (9993fea0...)
5/3/2026, 7:53:57 PM	Eray Tüzün COORDINATOR	LOGIN SUCCESS	User (8f576aea...)
5/3/2026, 7:52:27 PM	İsimsiz kullanıcı LABMEMBER	LOGIN SUCCESS	User (d0f0538a...)
5/3/2026, 7:51:50 PM	Semih Çağlar LABMEMBER	LOGIN SUCCESS	User (9993fea0...)

The Audit Logs tab shows recent system actions such as user logins, including the time, actor, action type, and related details. The administrator can filter logs by action type or date range and sort them by newest or oldest entries.

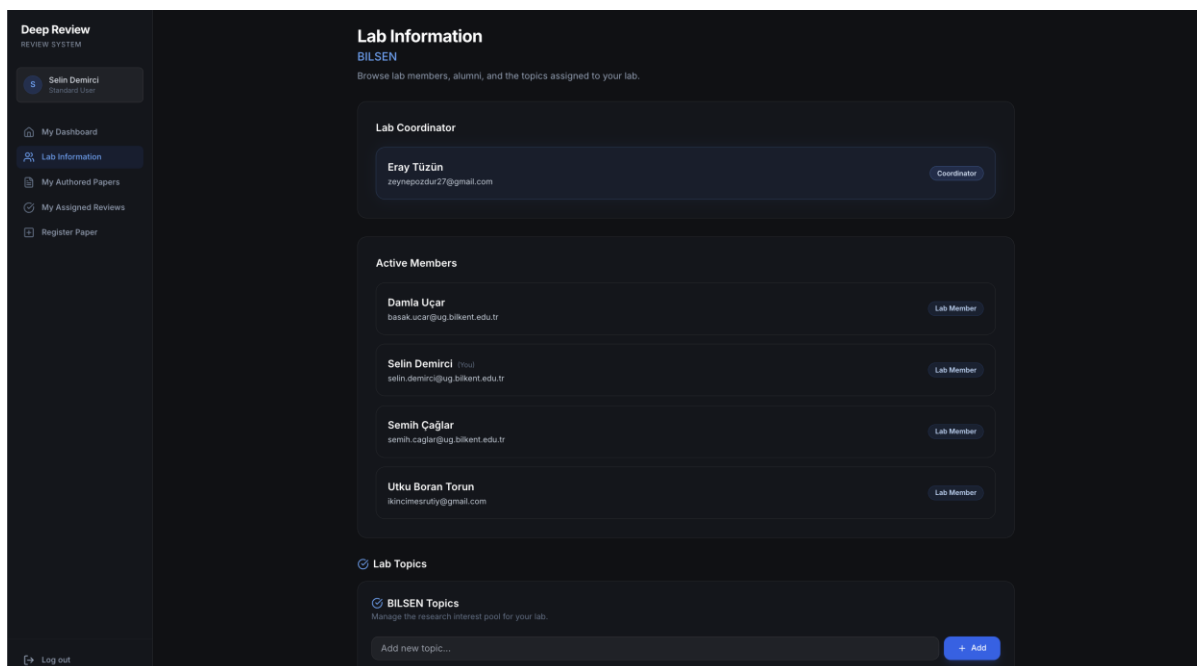
Lab Member Manual

Lab Member Dashboard – My Dashboard

The screenshot shows the 'My Dashboard' for a lab member named Selin Demirci. The dashboard includes a sidebar with 'My Dashboard', 'Lab Information', 'My Authored Papers', 'My Assigned Reviews', and 'Register Paper'. The main content area is titled 'Welcome, Selin Demirci' and features three summary cards: 'Account Status Active', 'Assigned Reviews 1', and 'Waiting Ratings 1'. Below these are two sections: 'Assigned Reviews' showing a review for 'SERSEM: Selective Entropy-Weighted Scoring for Membership Inference in Code Language Models' and 'Responses to Requests' showing a response for 'Better Automated Classification of Code Review Comments with Large Language Models'. The sidebar also includes a 'Log out' button.

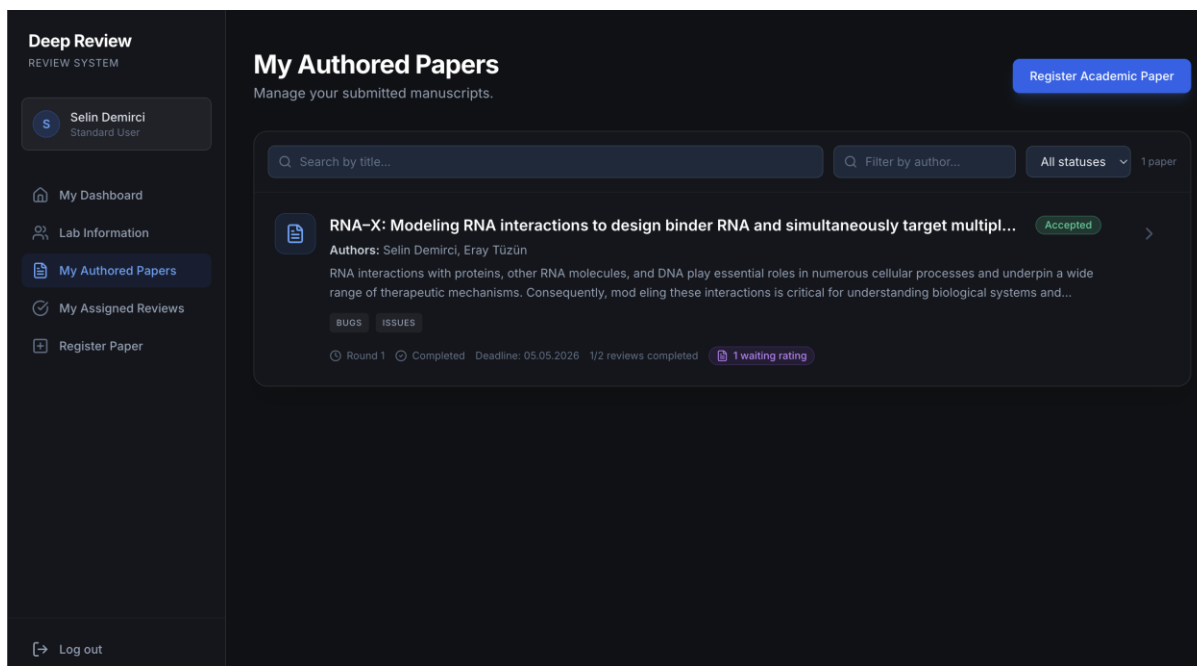
The My Dashboard page gives the lab member a summary of their account status, assigned reviews, and waiting ratings. It also shows quick sections for assigned reviews and responses to extension or decline requests.

Lab Member Information



The Lab Information page shows the lab coordinator, active lab members, and the topics assigned to the lab. Lab members can view their lab community and add new research topics from the Lab Topics section.

Lab Member – My Authored Papers



The My Authored Papers page lists the papers submitted by the lab member and shows each paper's status, review round, deadline, and rating status. Users can search or filter their papers and start a new submission by clicking **Register Academic Paper**.

Lab Member- Paper Detail Page

Deep Review
REVIEW SYSTEM

S

Selin Demirci
s@demirci.org

My Dashboard

Lab Information

My Authored Papers

My Assigned Reviews

Register Paper

Log out

← Back to Papers

RNA-X: Modeling RNA interactions to design binder RNA and simultaneously target multiple molecules of different types

ACCEPTED

Open Round Overview

bugs

issues

EDIT TOPICS

Open Overview

Abstract

RNA interactions with proteins, other RNA molecules, and DNA play essential roles in numerous cellular processes and underpin a wide range of therapeutic mechanisms. Consequently, modeling these interactions is critical for understanding biological systems and designing novel RNA-based therapeutics. However, designing RNA sequences that selectively and strongly bind to a specific target remains a major challenge due to the vast sequence space and the current limitations of experimental and computational methods. In this study, we introduce RNA-X, the first RNA interaction foundation model which is based on masked language modeling for representation learning, conditional RNA design, and optimization. For the first time, RNA-X enables (i) RNA design targeting not only proteins but also other RNA and DNA molecules, and (ii) simultaneous design against multiple targets. Our extensive experiments demonstrate that RNA-X generates RNA sequences with natural structural characteristics and surpasses state-of-the-art methods in protein targeting. Using affinity predictors and molecular dynamics simulations, we further show that the model can design RNA molecules that (i) target therapeutically relevant molecules such as p53 and thrombin proteins, and (ii) bind to targets with no prior interaction data, such as SDAD1 protein. As a proof of concept, we designed a novel guide RNA from scratch that simultaneously binds (i) to the DNA of a bacterium and (ii) to the Cas9 protein. The resulting design achieves a predicted binding energy comparable to that of the wild-type guide RNA reported in the Protein Data Bank. Despite having orders of magnitude fewer parameters than existing RNA foundation models, RNA-X produces representations that outperform them across diverse RNA interaction-related downstream tasks. The code and pretrained model are publicly available on GitHub

AUTHORS

S

Selin Demirci
s@demirci.org

E

Eray Tüzün
eraytuzun27@gmail.com

Round Overview

Open Round Overview

1

ACM Transactions on Computer Systems (TOCS)

Completed

Round Deadline

05.05.2026

History & Activity Log

ACCEPTED

RNA-X: Modeling RNA interactions to design binder RNA and simultaneously target multiple molecules of different types
ACM Transactions on Computer Systems (TOCS)

The Paper Detail page shows the selected paper's title, status, topics, abstract, authors, and related links. From this page, the user can edit paper information, open external paper links, view the round overview, and check the paper's history/activity log.

Round Overview

Deep Review
REVIEW SYSTEM

S

Selin Demirci
Standard User

My Dashboard

Lab Information

My Authored Papers

My Assigned Reviews

Register Paper

Log out

← Paper Overview

+ Create Next Round

Round Overview

RNA-X: Modeling RNA interactions to design binder RNA and simultaneously target multiple molecules of different types

1 round

Open Overleaf

1

ACM Transactions on Computer Systems (TOCS)

Journal | Round deadline: 05 May 2026

Completed

VENUE LINK

https://dl.acm.org/journal/tocs

ROUND DEADLINE

05 May 2026

AI Review Analyses

No saved AI analysis for this round yet.

AI Tools

Assigned Reviewers (2)

Submitted reviews show their summary and can be rated once.

D

Damla Uçar

basak.ucar@ug.bilkent.edu.tr

Declined

S

Semih Çağlar

semih.caglar@ug.bilkent.edu.tr

Completed

Rate Reviewer

REVIEW SUMMARY

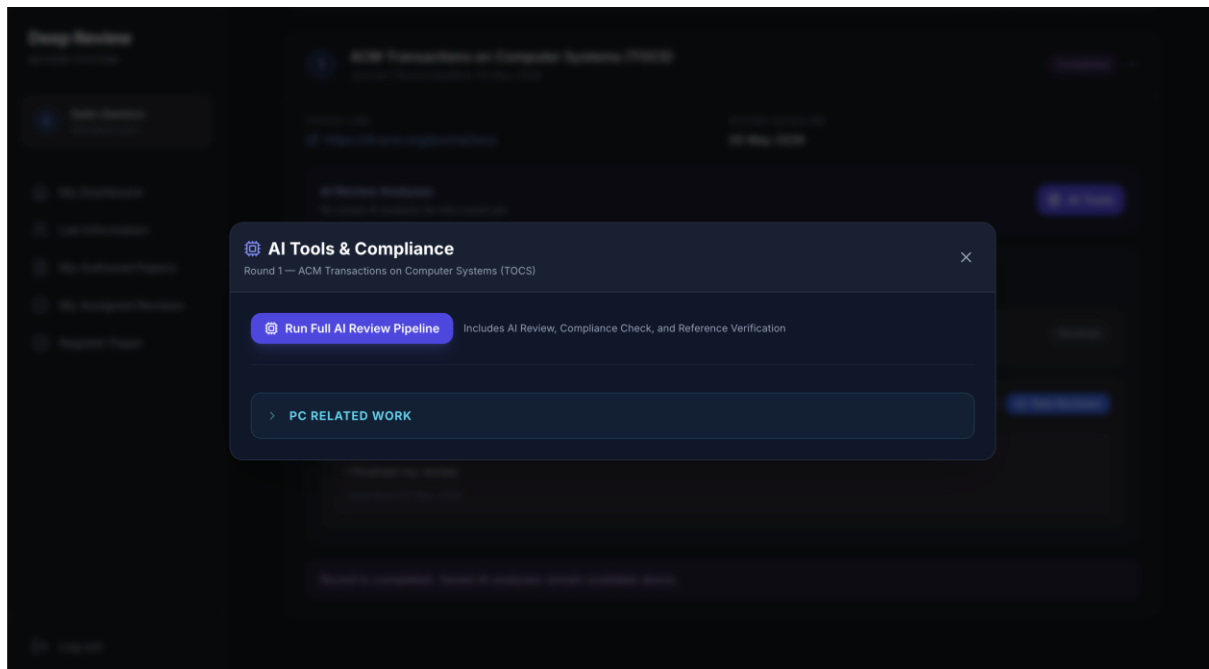
I finished my review

Submitted 03 May 2026

Round is completed. Saved AI analyses remain available above.

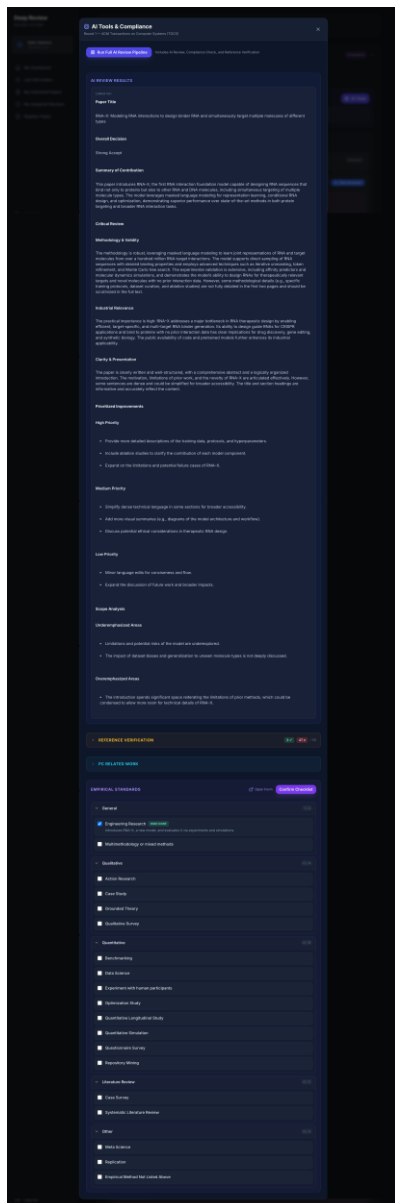
The Round Overview page shows the review round details for the selected paper, including the venue link, deadline, assigned reviewers, review status, and submitted review summaries. The user can open the paper link, access AI tools, rate completed reviewers, or create the next review round when available.

Lab Member – AI Tools & Compliance



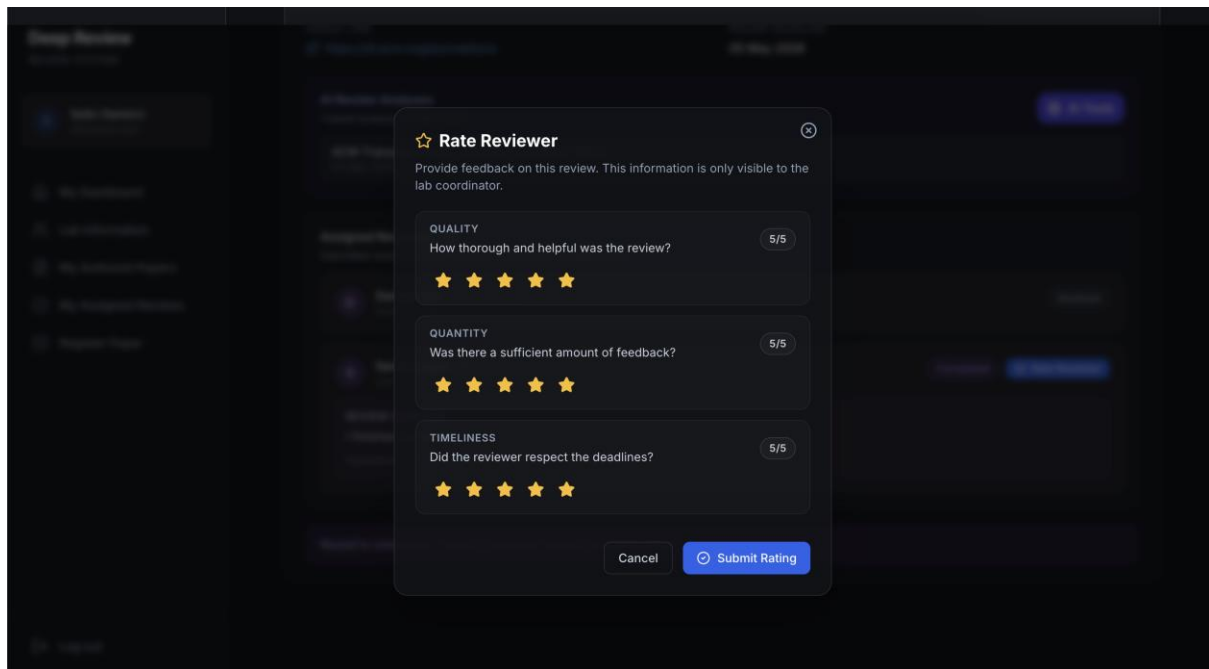
The AI Tools & Compliance window lets the user run the full AI review pipeline for the selected review round. It also provides access to PC related work analysis for checking related papers and compliance information.

Lab Member – AI Review Results and Checklist



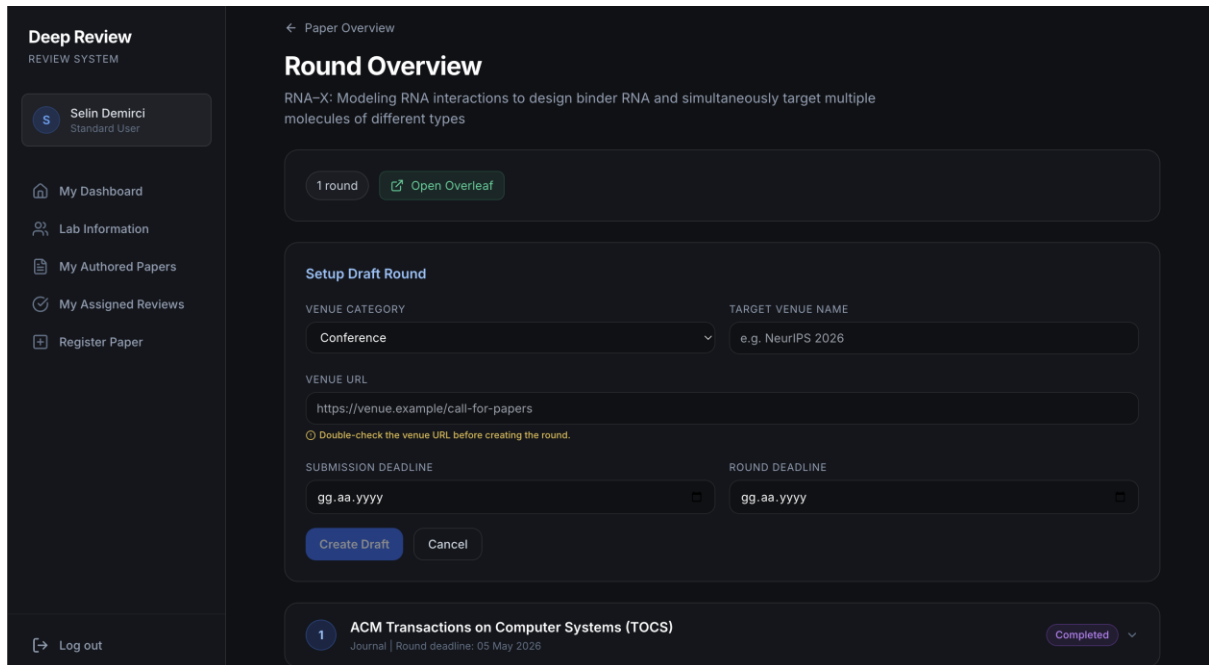
After running the AI review pipeline, the system displays an AI-generated review report with sections such as overall decision, contribution summary, critical review, and improvement priorities. The same window also includes reference verification, PC related work, and empirical standards checklists that help the user review the paper's compliance and completeness.

Lab Member – Rate Reviewer

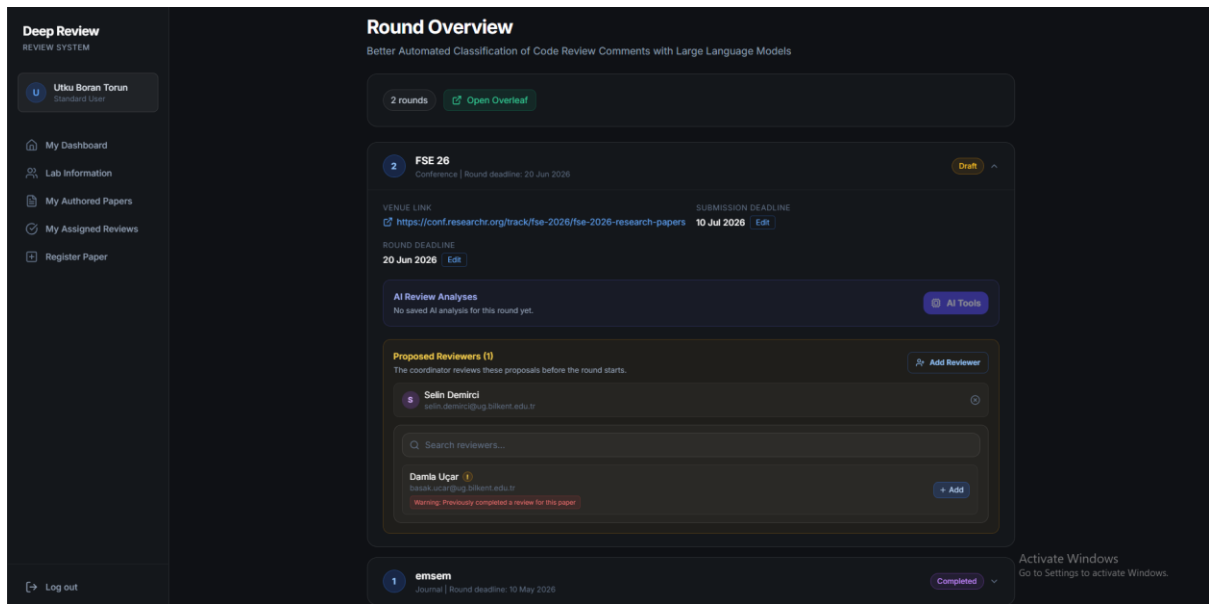


The Rate Reviewer window allows the author to evaluate a completed review using quality, quantity, and timeliness scores. After selecting the star ratings, the user can submit the rating, which is only visible to the lab coordinator.

Lab Member – Setup Draft Round

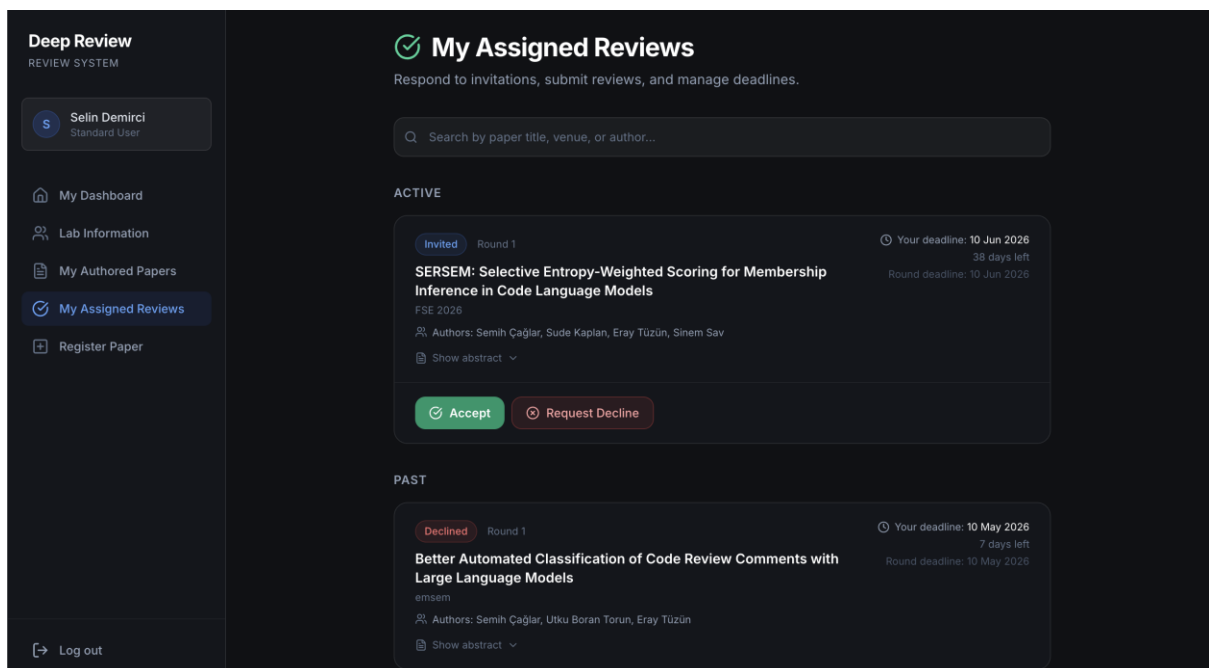


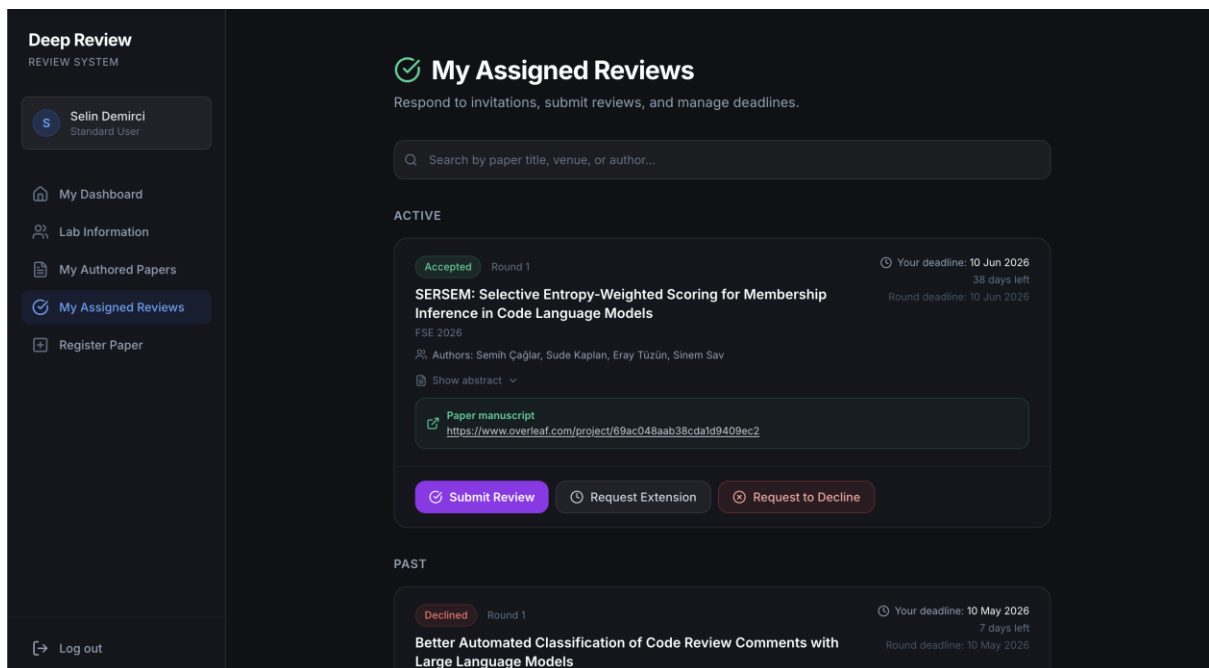
The Setup Draft Round section lets the author propose to create a new review round by entering the venue category, venue name, venue URL, submission deadline, and round deadline. The system reminds the user to double-check the venue URL before creating the draft round.



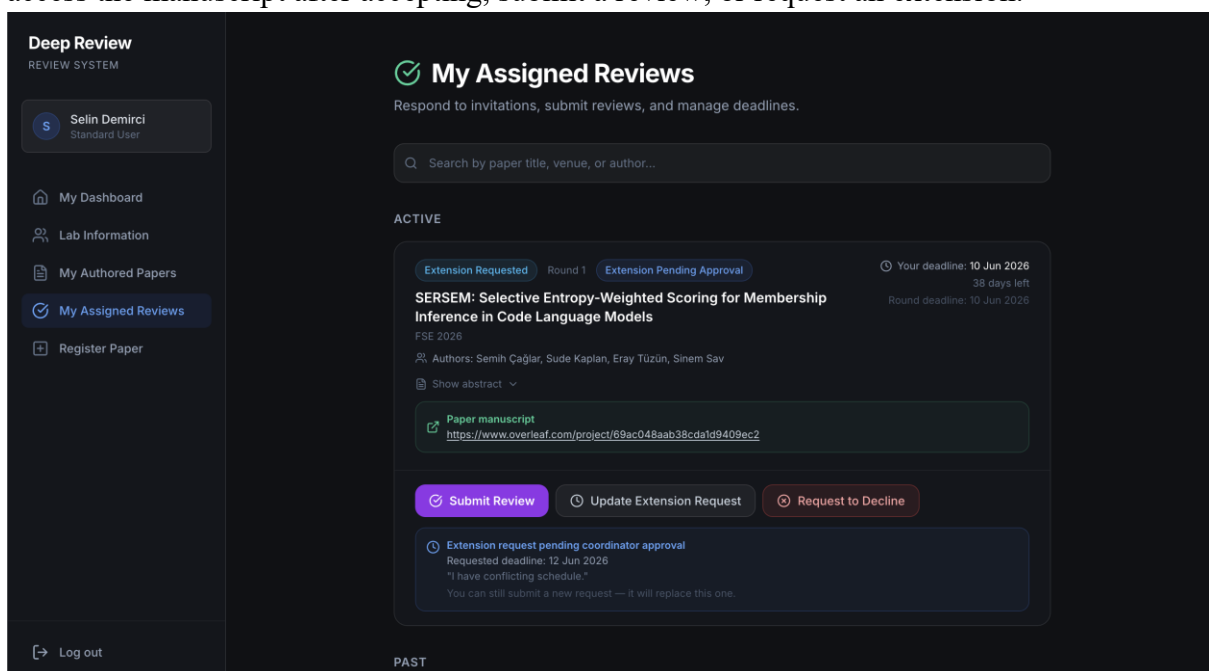
The user can view reviewer proposals and add eligible reviewers while the coordinator reviews the draft round.

Lab Member – My Assigned Reviews





The My Assigned Reviews page shows active and past review assignments with their deadlines, status, paper details, and available actions. Users can accept or decline invitations, access the manuscript after accepting, submit a review, or request an extension.



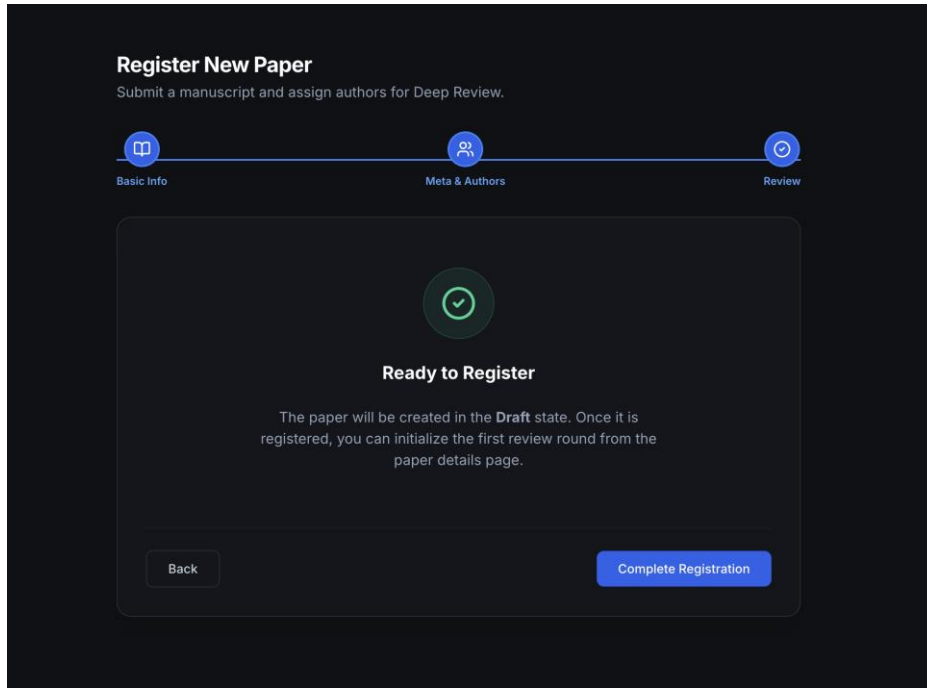
After requesting an extension, the assignment card shows the request as pending coordinator approval. The user can still submit the review, update the extension request, or request to decline the assignment.

Lab Member – Register New Paper

The screenshot shows the 'Register New Paper' interface in the 'Basic Info' step. The left sidebar contains the user's name 'Selin Demirci' and a list of navigation items: 'My Dashboard', 'Lab Information', 'My Authored Papers', 'My Assigned Reviews', and 'Register Paper'. The main content area has a progress bar with three steps: 'Basic Info' (active), 'Meta & Authors', and 'Review'. Below the progress bar, the 'Paper Title' field contains the placeholder 'Enter full paper title...'. The 'Abstract' field contains the placeholder 'Provide a detailed abstract of the work...'. The 'Overleaf Link' field contains the placeholder 'https://www.overleaf.com/...'. A 'Continue' button is located at the bottom right.

The screenshot shows the 'Register New Paper' interface in the 'Meta & Authors' step. The left sidebar is identical to the previous screenshot. The progress bar now highlights the 'Meta & Authors' step. The 'Assign Authors' section has a search bar 'Search by name or email...' and three author cards: 'Damla Uçar' (damla.ucar@bug.bilkent.edu.tr), 'Semih Çağlar' (semih.caglar@bug.bilkent.edu.tr), and 'Utku Boran Torun' (utkuboran19@gmail.com). The 'Author List & Order' section shows a list of authors: '1. Selin Demirci' and '2. Eray Tüzün' (Coordinator), with up and down arrows for reordering. The 'Topics & Keywords' section has a search bar 'Search topics...' and four topic buttons: 'bugs', 'issues', 'LLMs', and 'code review'. A 'Back' button is at the bottom left and a 'Continue' button is at the bottom right.

The Register New Paper page lets the lab member submit a new manuscript by entering the paper title, abstract, and Overleaf link. The submission process is organized into steps for basic information, metadata/authors, and final review. In the Meta & Authors step, the user assigns authors to the paper, reviews the author order, and selects relevant topics or keywords. These topics help the system match the paper with suitable reviewers.



Coordinator – My Dashboard

Deep Review
REVIEW SYSTEM

E Eray Tüzün
Coordinator

My Dashboard

Lab Information

Pending Approvals

Papers

Register Paper

Round Management

Log out

Welcome, Eray Tüzün

Review pending account requests and keep the system moving.

Pending Approvals
0

Round Requests
1

Collaboration Invitations
0

Reviewer Metrics

Performance, assignment volume, acceptance/rejection, and timeliness are tracked per reviewer for the selected period.

Monthly Yearly Overall

Search reviewer by name...

#	REVIEWER	AGGREGATE	QUALITY	QUANTITY	TIMELINESS	ASSIGNED	COMPLETED	INCOMPLETE
1	Damla Uçar basak.ucar@ug.bilkent.edu.tr	5.00	5.00	5.00	5.00	2	1	0
2	Selin Demirci selin.demirci@ug.bilkent.edu.tr	-	-	-	-	2	0	0
3	Semih Çağlar semih.caglar@ug.bilkent.edu.tr	-	-	-	-	1	1	0
4	Utku Boran Torun ikincimesrutiy@gmail.com	-	-	-	-	0	0	0

Pending Requests

Round start Round 2 Open

Better Automated Classification of Code Review Comments with Larg...
Blisen - Journal - due 13 May 2026

Reviewer Requests

Decline request Round 1 Open

SERSEM: Selective Entropy-Weighted Scoring for Membership Inference...
Selin Demirci: I have other things to do

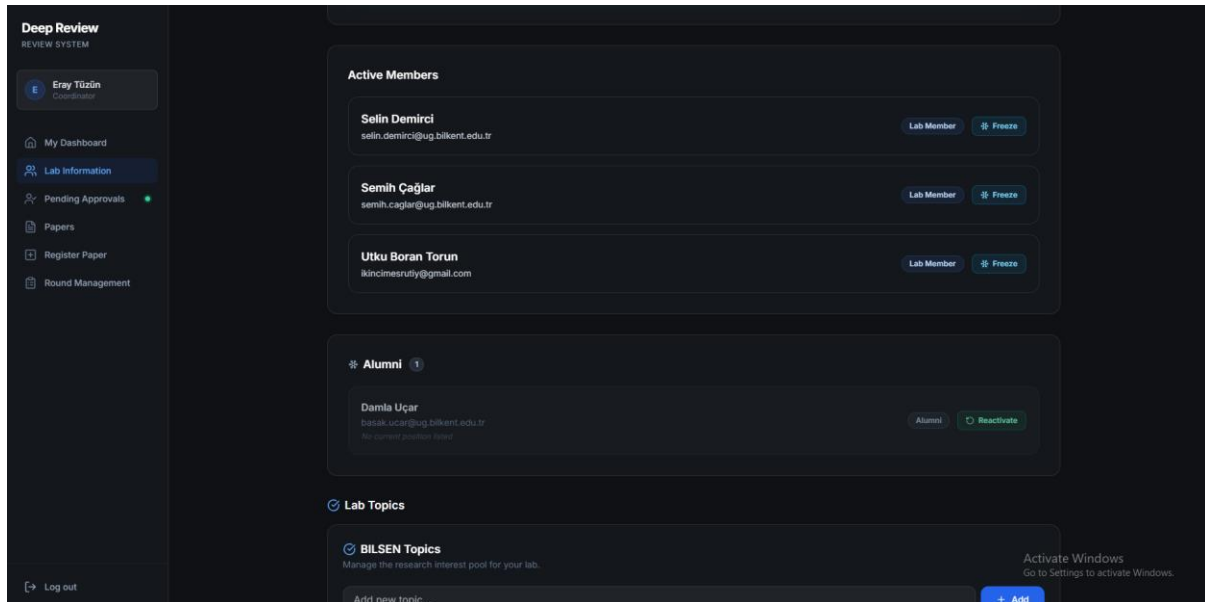
Extension request Round 1 Open

SERSEM: Selective Entropy-Weighted Scoring for Membership Inference...
Selin Demirci: I have conflicting schedule.

The coordinator dashboard summarizes pending approvals, round requests, and collaboration

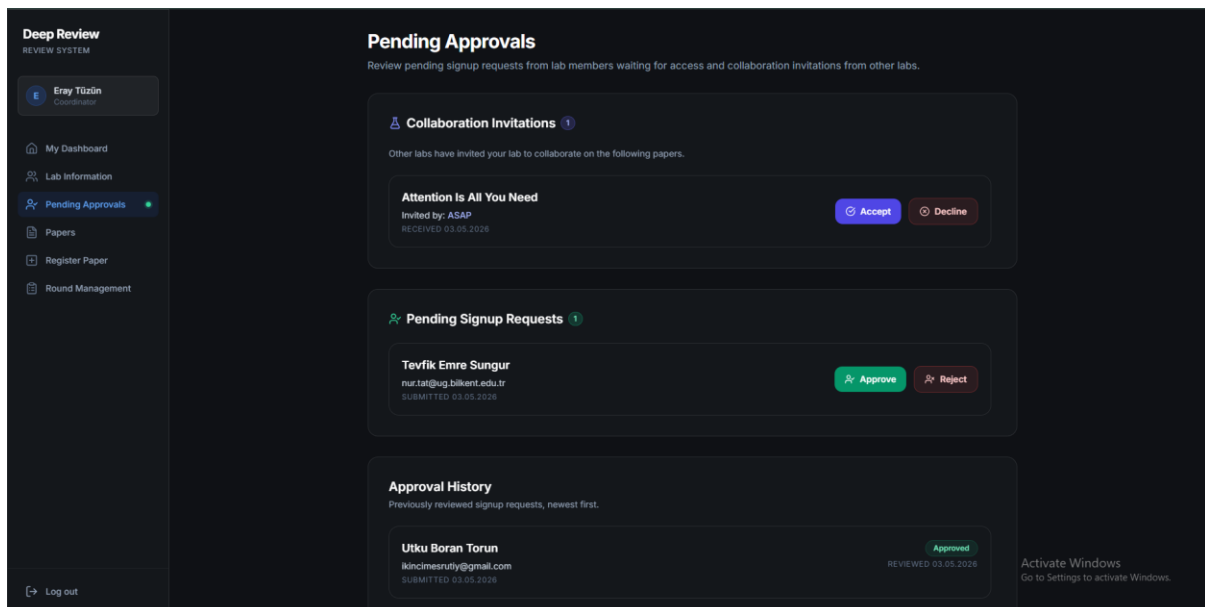
invitations. It also shows reviewer metrics and pending reviewer requests so the coordinator can monitor review activity and respond to requests quickly.

Coordinator – Lab Information



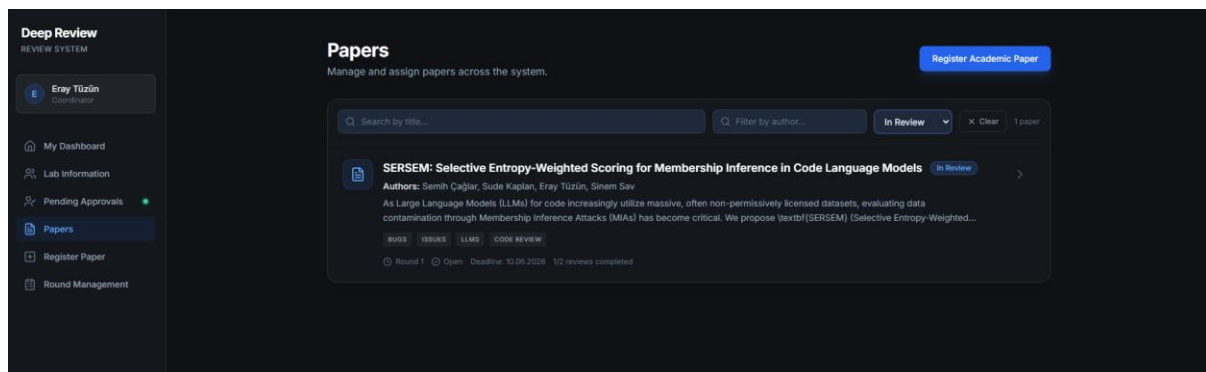
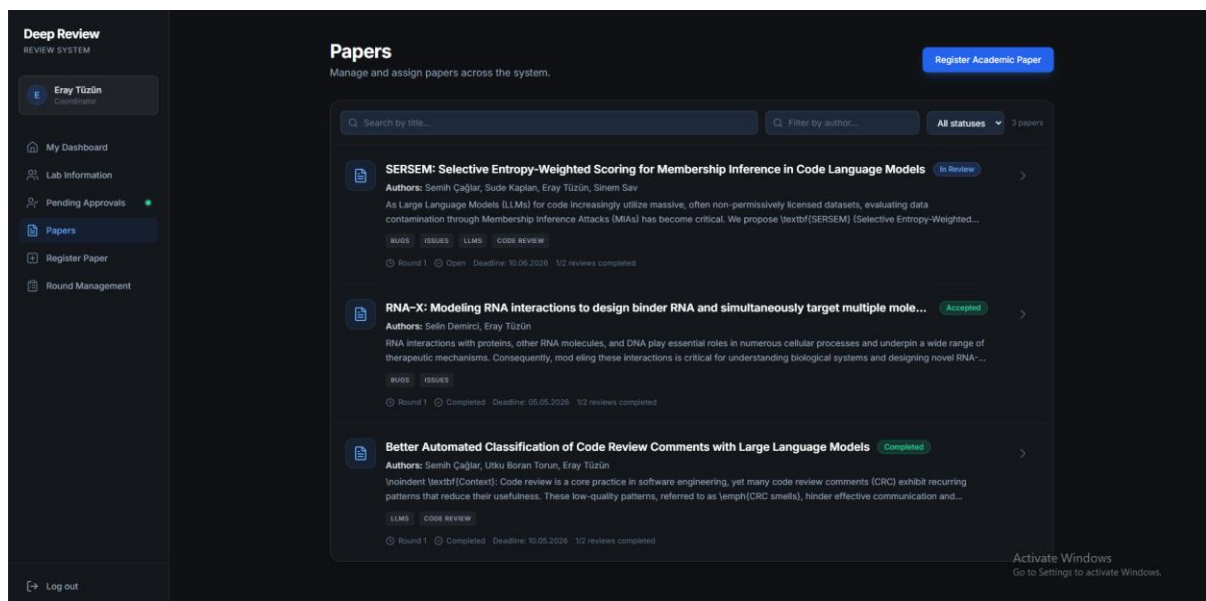
The Lab Information page lets the coordinator view active lab members, alumni, and lab topics. The coordinator can freeze active members, reactivate alumni, and add new topics to the lab's topic list.

Coordinator – Pending Approvals



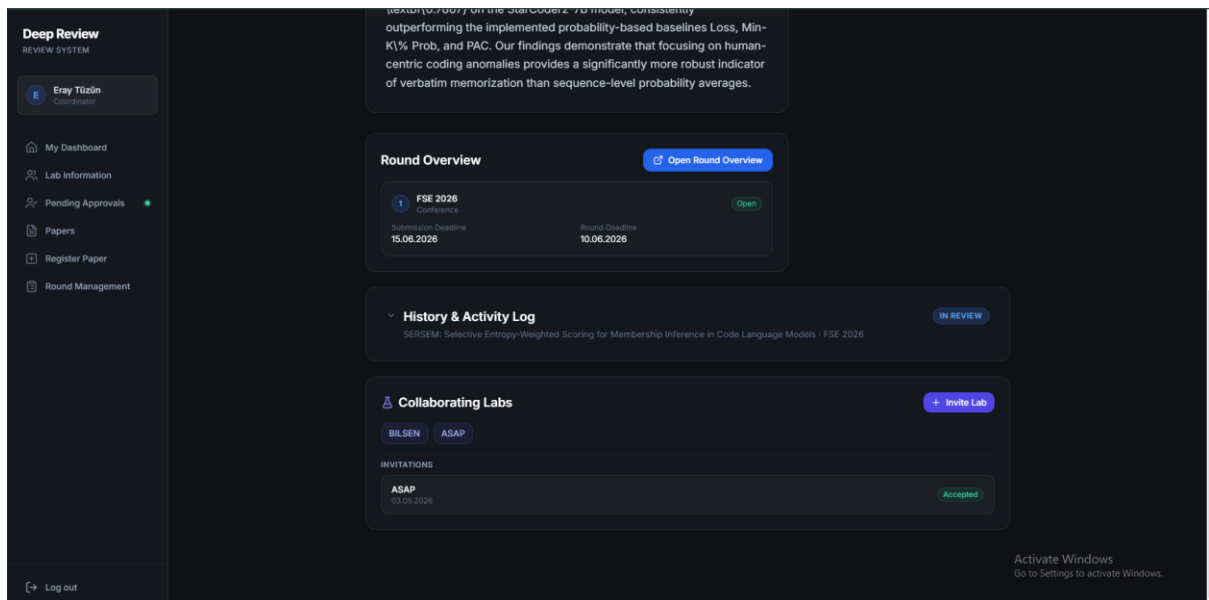
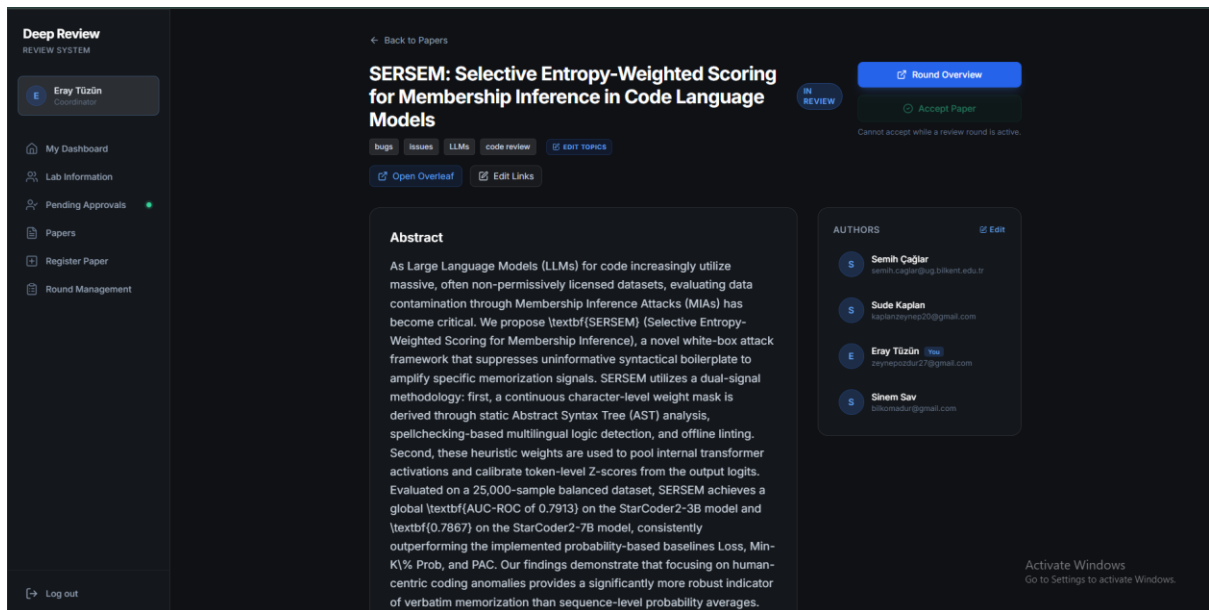
The Pending Approvals page lets the coordinator review collaboration invitations and new signup requests. The coordinator can accept or decline collaboration invitations, approve or reject user registrations, and view previous approval decisions.

Coordinator – Papers

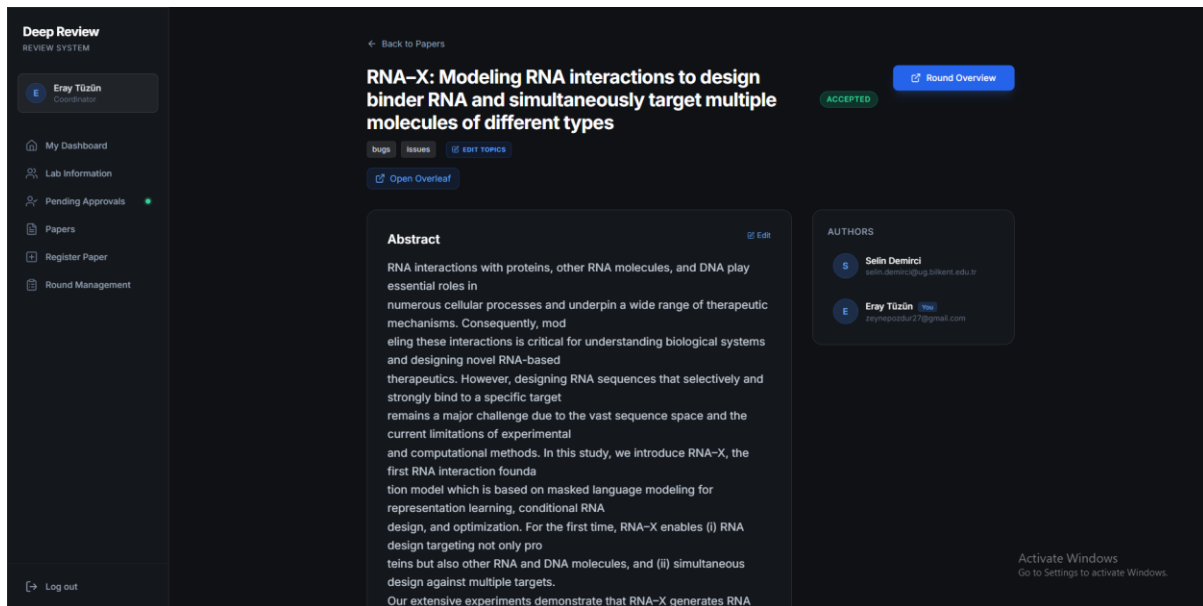


The Papers page shows all papers that the coordinator can manage, including their authors, status, topics, deadlines, and review progress. The coordinator can search or filter papers, open paper details, and register a new academic paper when needed.

Coordinator – Paper Detail Page

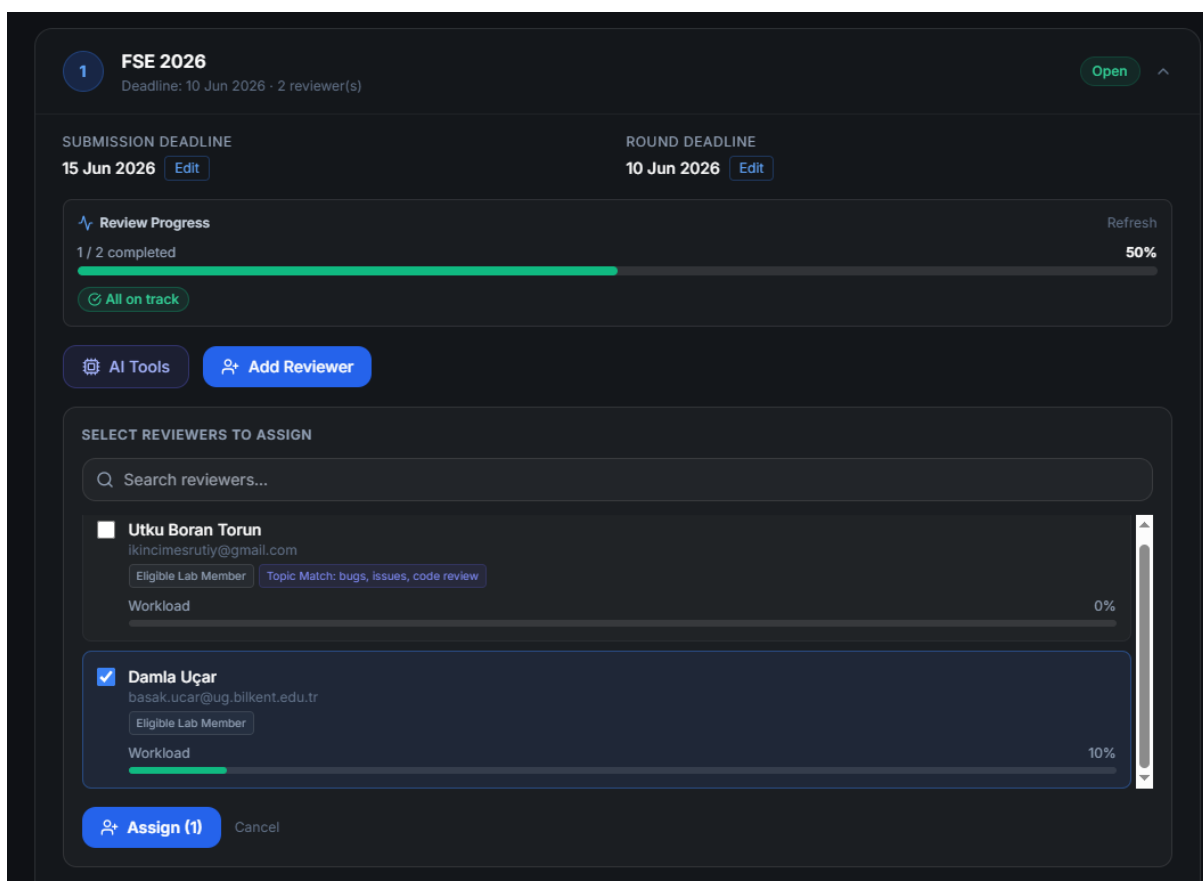


The Paper Detail page lets the coordinator view and manage a selected paper's abstract, authors, topics, links, round overview, and activity history. The coordinator can also invite collaborating labs and monitor whether collaboration invitations are pending or accepted.



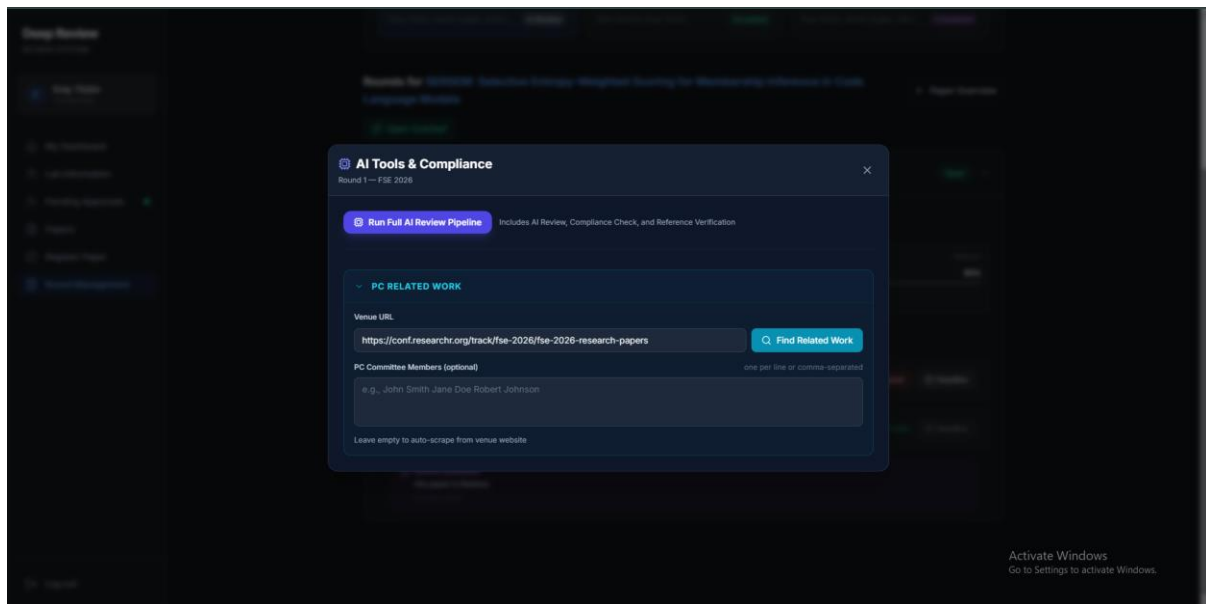
After a paper is accepted, new round for that paper cannot be started.

Coordinator – Add Reviewer



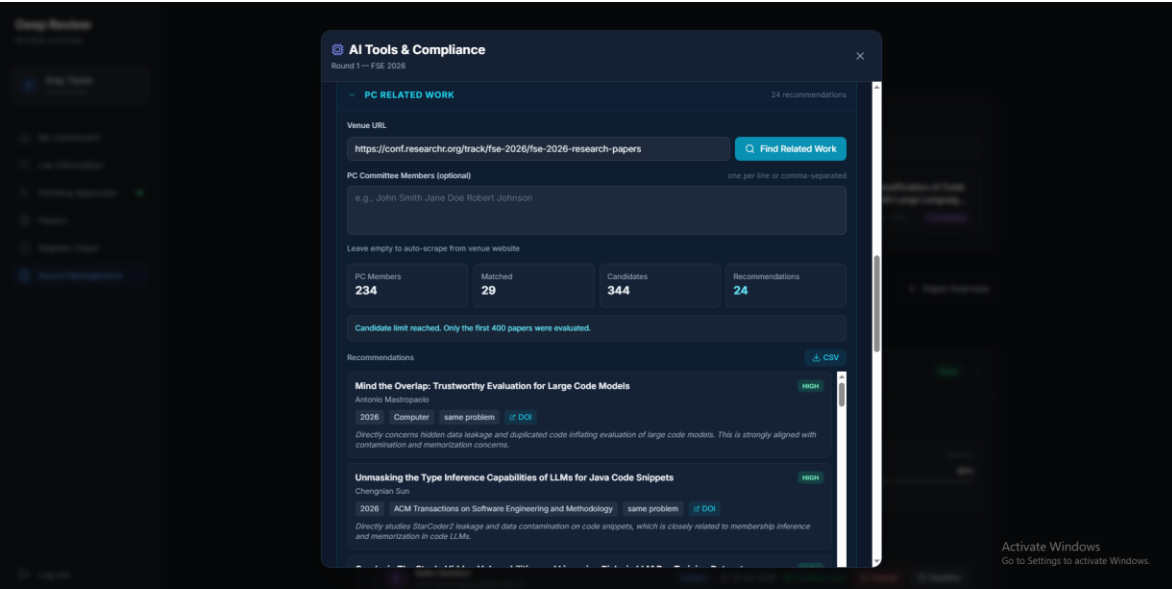
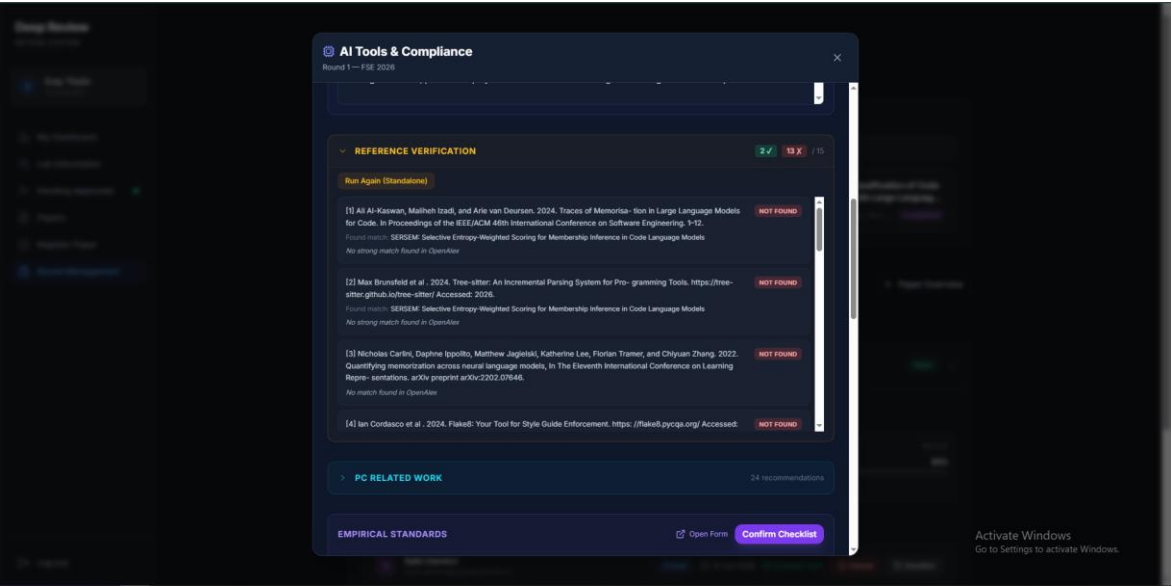
The Add Reviewer section lets the coordinator search and select eligible lab members for a review round. The coordinator can compare reviewer topic matches and workload, then click **Assign** to add the selected reviewer.

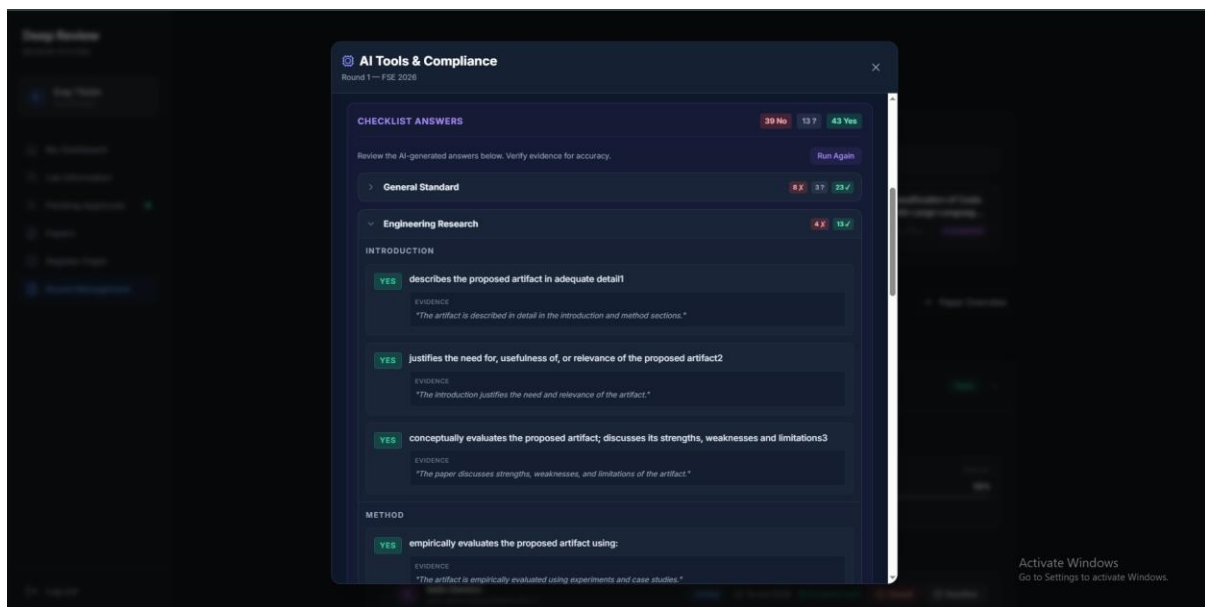
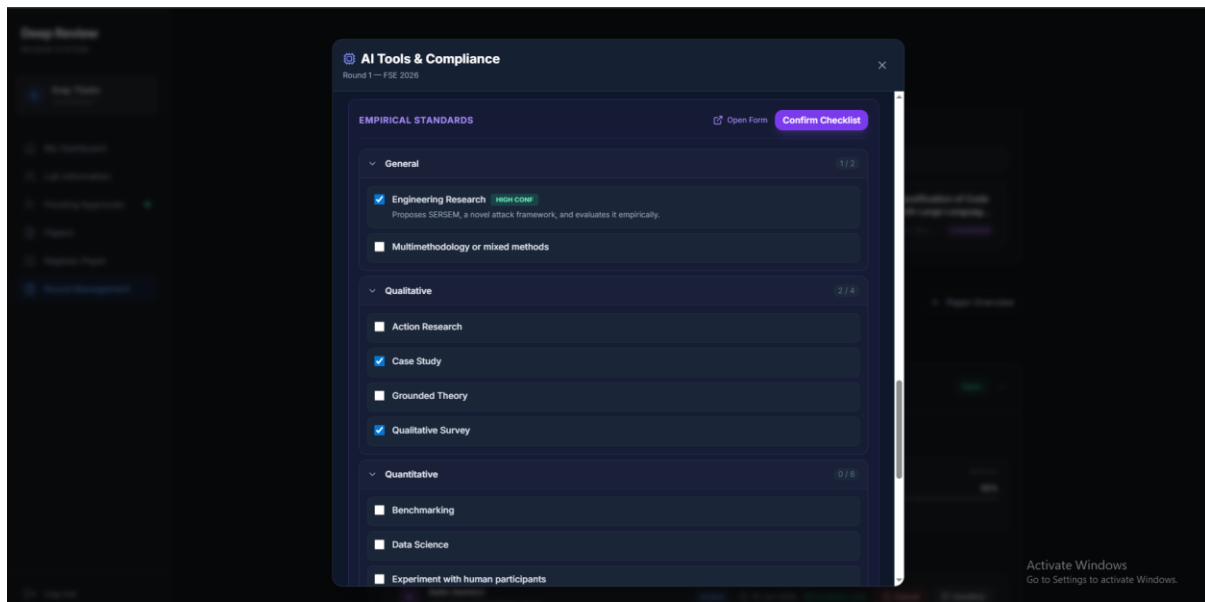
Coordinator – PC Related Work



The PC Related Work section helps the coordinator find related work for the selected review round. The coordinator can enter a venue URL and optional PC committee member names, then run the related work search.

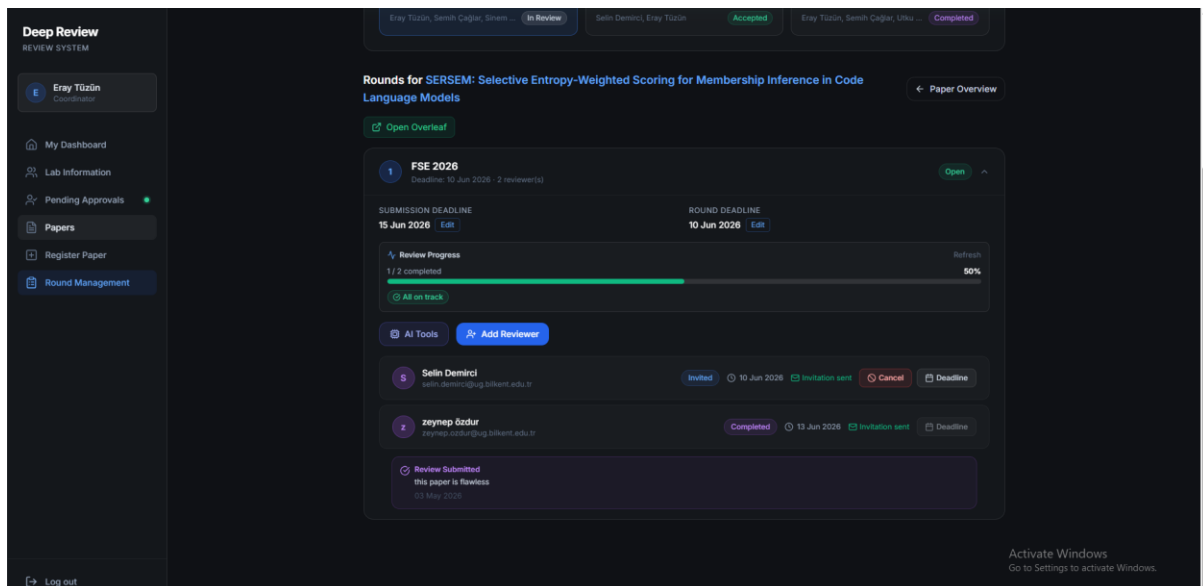
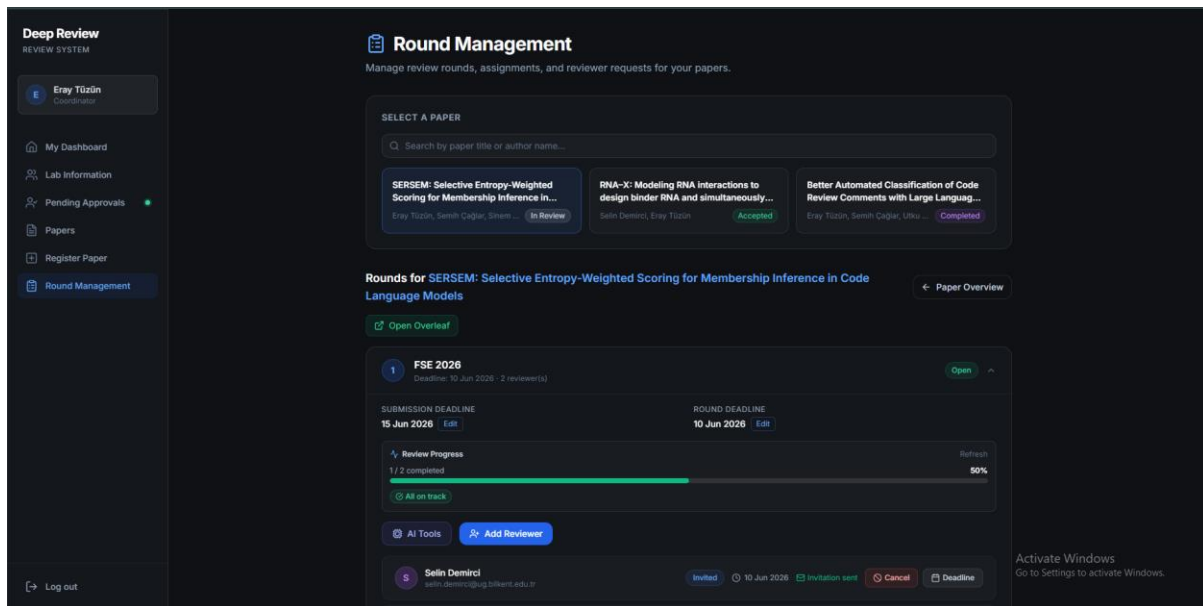
Coordinator – AI Tools Results





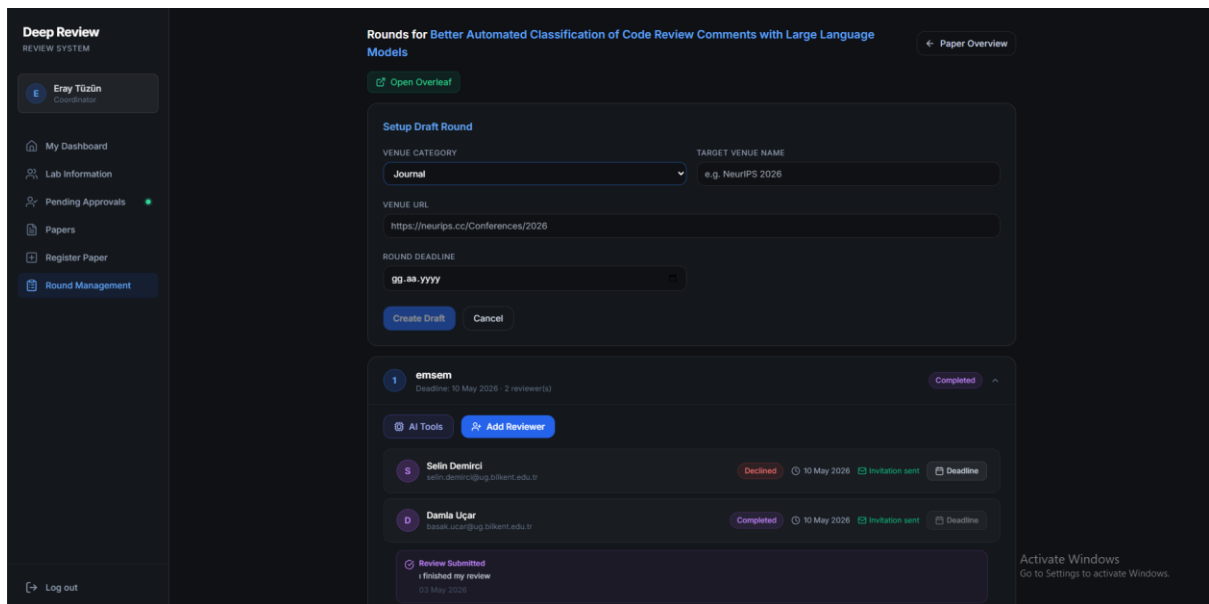
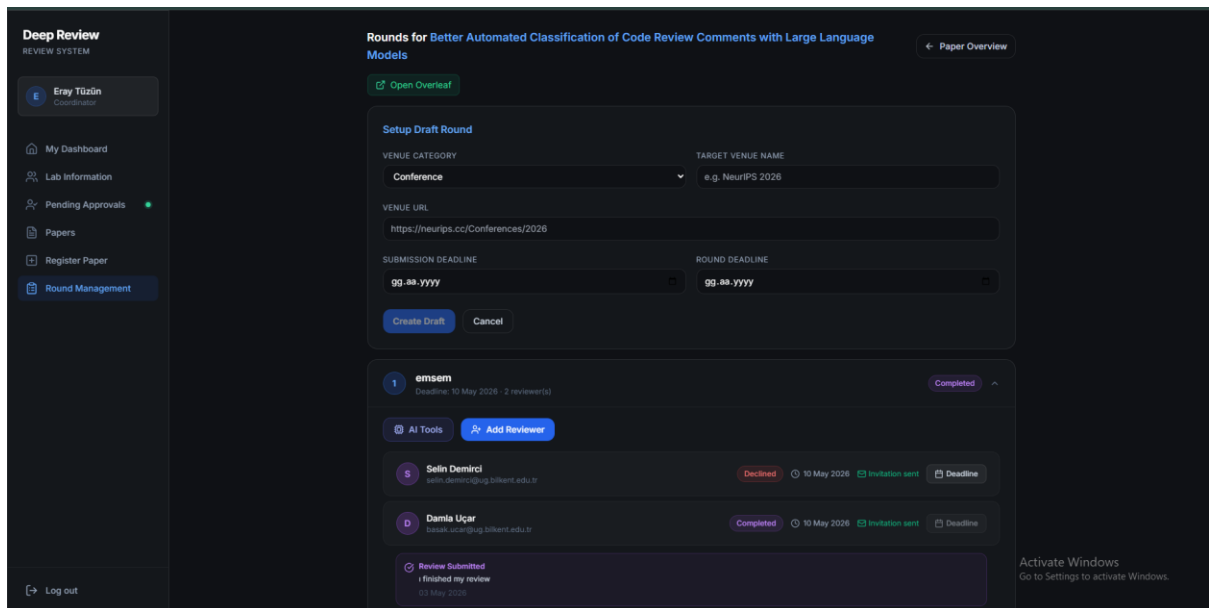
The AI Tools & Compliance window shows reference verification results, related work recommendations, and empirical standards checks for the selected round. The coordinator can review the AI-generated findings, confirm the checklist, and inspect the evidence behind each checklist answer.

Coordinator – Round Management



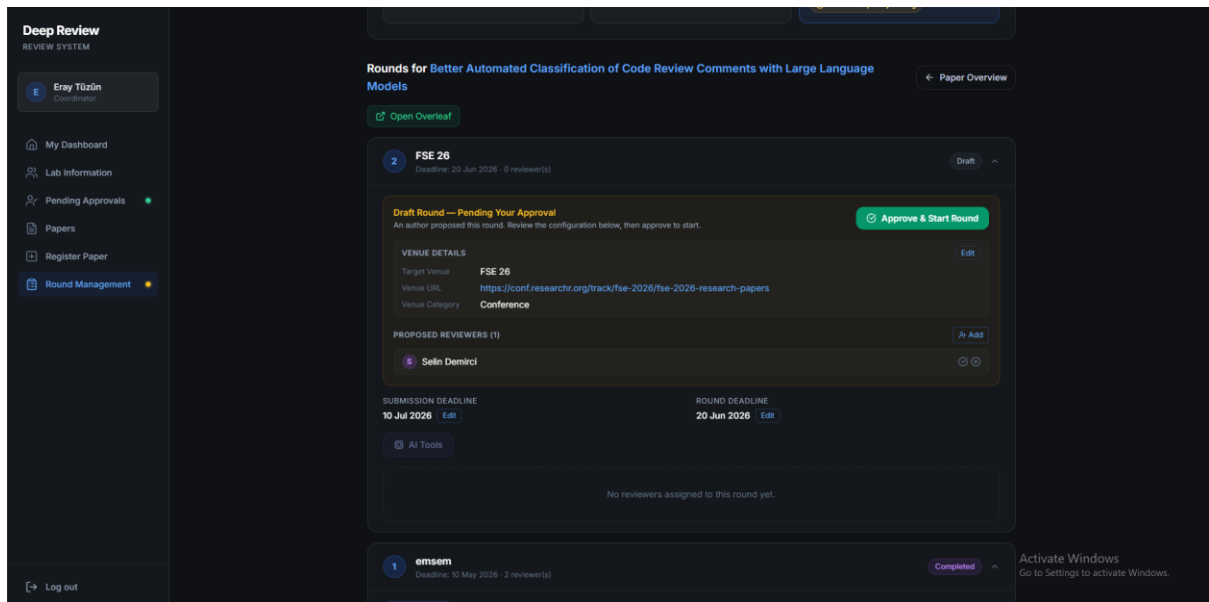
The Review Round Details section shows the selected paper's round deadline, review progress, reviewer list, invitation status, and submitted reviews. The coordinator can add reviewers, access AI tools, cancel invitations, or update reviewer deadlines from this page.

Coordinator – Setup Draft Round



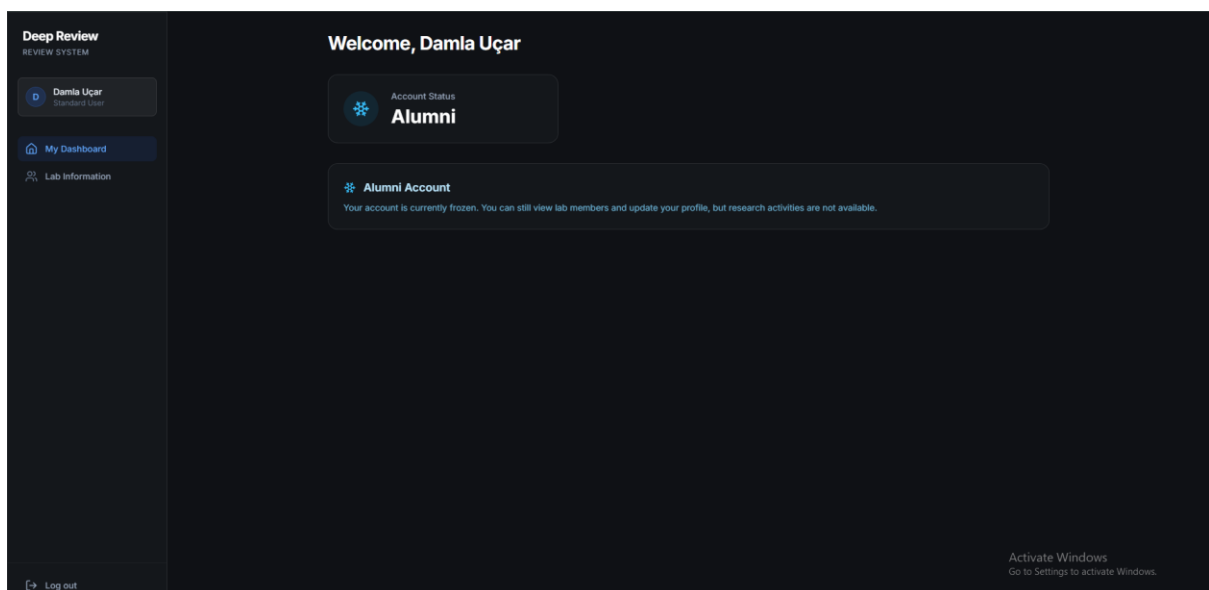
The Setup Draft Round section allows the coordinator to create another review round for a selected paper. The coordinator chooses the venue category, enters the target venue name, venue URL, and required deadlines, then clicks **Create Draft**.

Coordinator – Draft Round Approval

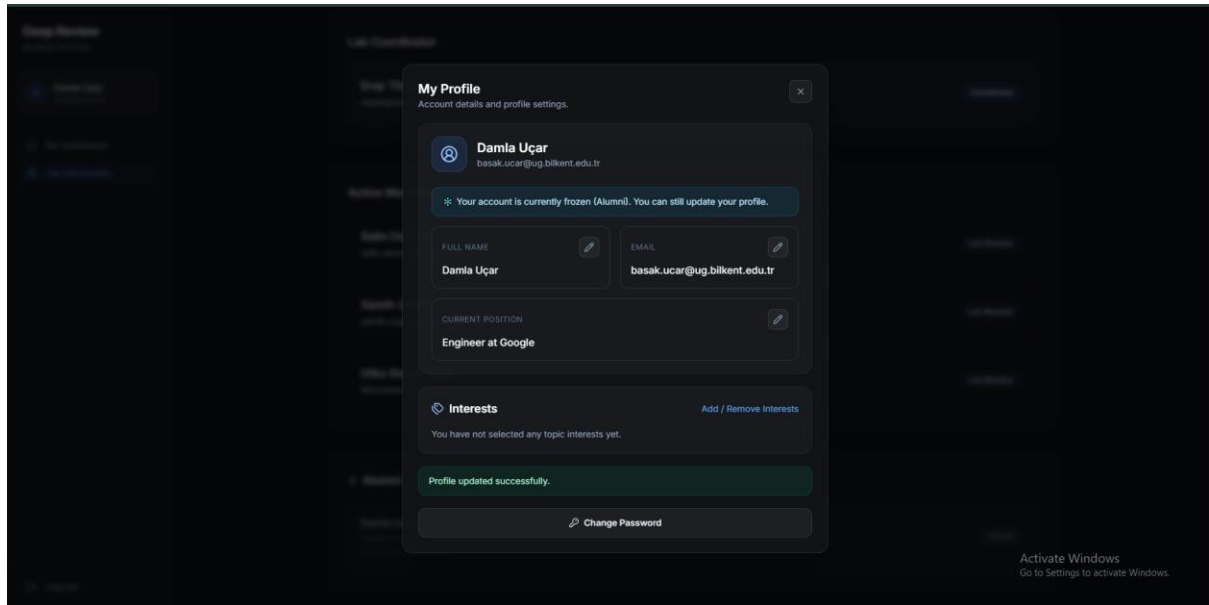


When an author proposes a draft review round, the coordinator can review the venue details, proposed reviewers, and deadlines before approving it. After checking the configuration, the coordinator can click **Approve & Start Round** to begin the review process.

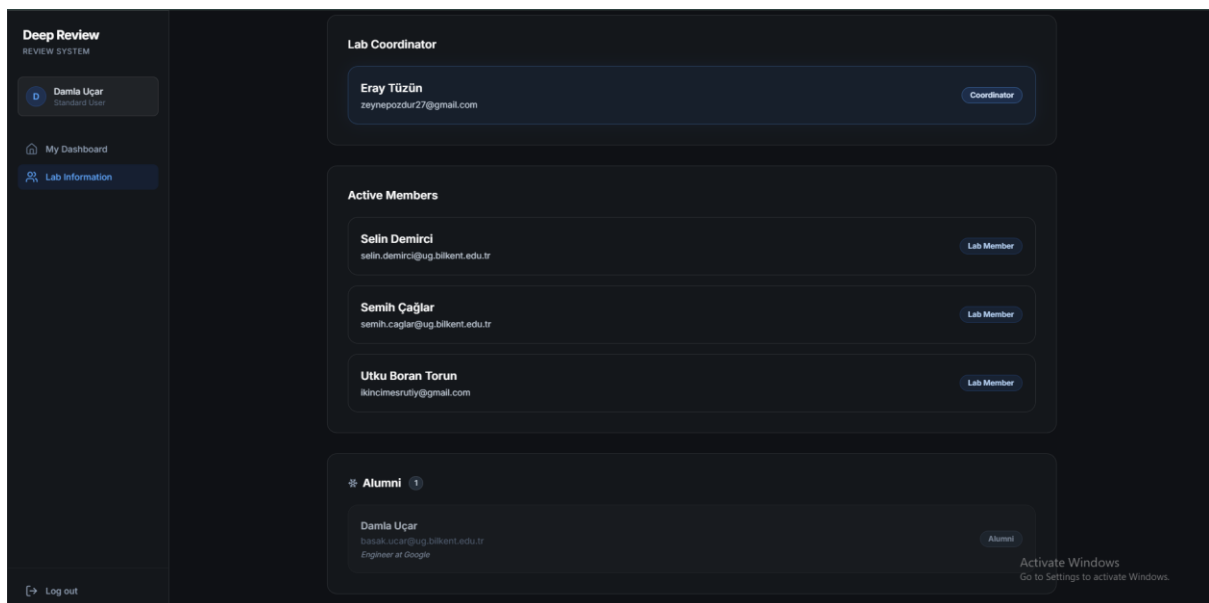
Alumni Dashboard View



The My Dashboard page for an alumni user displays their frozen account status. It notifies the user that while they retain access to view lab information and update their personal profile, they are restricted from participating in any active research activities.



The My Profile modal allows users to view and update their personal account details, such as their name, email, current position, and research interests. Even users with a frozen alumni status can access this menu to keep their information up to date, change their password, and receive on-screen confirmation when their profile modifications are saved successfully.



The Lab Information page provides a comprehensive directory of everyone associated with the user's lab. It is organized into distinct sections to display the Lab Coordinator, a list of Active Members currently participating in research, and an Alumni section for former members whose accounts are frozen but remain part of the lab's historical record.

8. Work Allocation

- **Damla Başak Uçar**
 - Worked on authentication, account, and profile-related workflows.
 - Implemented and refined signup/login, account approval, password reset, and change-password flows.
 - Contributed to profile editing, interests, profile modal behavior, account security, and related test/assumption documentation.
- **Zeynep Özdemir**
 - Worked on paper registration and paper management workflows.
 - Implemented paper creation, editing, field validation, author/topic selection, and Overleaf link handling.
 - Contributed to paper status updates, registration UI improvements, paper detail behavior, and related integration fixes.
- **Esra Nur Tat**
 - Worked on the review round and assignment lifecycle.
 - Implemented round creation, deadline handling, reviewer assignment/reassignment, reminders, and overdue support.
 - Contributed to reviewer invitation flow, lab collaboration invitations, email notifications, and related backend/integration tests.
- **Selin Demirci**
 - Worked on reviewer-facing workflows and review-management user experience.
 - Implemented invitation accept/decline behavior, extension and decline requests, review completion, and reviewer dashboard improvements.
 - Contributed to paper history, confirmation flows, round-management UX, reviewer reuse safeguards, and related frontend/backend fixes.
- **Semih Çağlar**
 - Worked on AI-assisted review, checklist, compliance, and reference-verification functionality.
 - Implemented PDF-based AI review support, AI review history, annotated PDF output, empirical checklist support, and reference checking.
 - Contributed to AI backend services, round/paper AI integrations, AI-related API routes, and frontend display of AI outputs.
- **Sude Kaplan**
 - Worked on administration, coordinator/lab management, ratings, and analytics.
 - Implemented admin UI improvements, user/lab/topic management, coordinator creation, and role-management refinements.
 - Contributed to reviewer rating, analytics support, audit/policy/template entities, documentation, and integration fixes.